
EVQ-Cosh: Variational Frequency Allocation for Rotary Position Embedding

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 The default RoPE frequency table is not the RoPE operator; it is an inherited al-
2 location of a scarce spectral budget. We study training-time frequency allocation
3 as a RoPE design axis complementary to operator design and inference-time scal-
4 ing. From a stated broadband collision surrogate over log-frequency densities, we
5 derive EVQ-COSH: a closed-form inverse-CDF allocation for the same finite set
6 of RoPE channels. It changes only the inverse-frequency initialization, adds zero
7 learned parameters, and uses $\tau = d_{\text{head}}/\sqrt{L}$ (or architecture-specific $d_{\text{eff}}/\sqrt{L}$)
8 as an operating default inside an empirical basin rather than as a claimed global
9 optimum. Mechanism stress tests show that allocation shape changes how range
10 scaling and compressed-RoPE architectures use their channels: under matched
11 YaRN scale on a 454M passkey-mix model, EVQ+YaRN reaches 100% teacher-
12 forced NLL-gap passkey retrieval at 8K while Geo+YaRN remains at 61% across
13 three seeds; in a 432M MLA model with only 16 rotary channels, EVQ alone
14 reduces $2\times$ extrapolation PPL by 31.1% at +1.1% in-distribution cost. Together,
15 the primary stress tests indicate that the geometric schedule is not a neutral imple-
16 mentation detail, but a substantive inductive bias for long-context behavior.

17 1 Introduction

18 Rotary position embedding (RoPE; [Su et al., 2024](#)) is often discussed as a positional operator, but in
19 implementation it is also a *finite spectral allocation*: each attention head receives only $d_{\text{head}}/2$ rotary
20 frequency channels, and the default geometric (equal-ratio) schedule is one inherited convention for
21 spending this budget. This equal-ratio convention remains common across LLaMA-3 [[Grattafiori](#)
22 [et al., 2024](#)], Qwen2.5 [[Yang et al., 2024a](#)], DeepSeek-V3 [[DeepSeek-AI, 2024b](#)], and video DiTs
23 [[Wei et al., 2025](#)], modulo base choices, axis handling, and architecture-specific compression. Once
24 RoPE is viewed as a finite spectral budget, a basic question appears: before training begins, how
25 should these channels be allocated?

26 The rotation is the operator; the frequency table is a design choice.

27 Most long-context work answers a different question. Base-frequency tuning changes the span of
28 the spectrum; positional interpolation [[Chen et al., 2023](#)], NTK-aware scaling [[Rozière et al., 2023](#)],
29 YaRN [[Peng et al., 2024](#)], and LongRoPE-style methods [[Ding et al., 2024](#), [Shang et al., 2025](#)]
30 extend usable range after the allocation family has been fixed, via interpolation, scaling, search,
31 or adaptation. These *range-scaling* methods stretch or reuse a frequency table whose allocation
32 was fixed at initialization. Adjacent works observe that the allocation matters empirically—[Barbero](#)
33 [et al. \[2025\]](#) find high- and low-frequency specialization after training, [Wei et al. \[2025\]](#) propose
34 a modality-specific allocation for video, [Wang et al. \[2024\]](#) preserve critical frequencies during
35 interpolation—but, to our knowledge, none gives a closed-form *training-time* allocation for standard
36 RoPE derived from an explicit variational surrogate.

We treat training-time frequency-channel allocation as a third RoPE design axis, complementary to operator and range. Parameterizing allocations by densities $\rho(\phi)$ over log-frequency, we introduce EVQ-COSH: a closed-form one-parameter allocation family with zero learned parameters that changes only the RoPE inverse-frequency initialization. Geometric RoPE is recovered as the $\tau \rightarrow 0$ limit of the pure-tether sub-family (not a generic optimum). A small- τ softmax-transport calculation motivates the deployable operating rule $\tau = d_{\text{head}} / \sqrt{L}$, used as a robust default inside an empirically flat basin rather than a claim of global optimum.

Empirical scope. We report three *primary* stress tests (Primary I–III in §4, 3-seed where indicated): a matched-scale substrate test composing EVQ with YaRN on a 454M transformer (3-seed); a PE-dominant 128→8K extrapolation test against DAPE [Zheng et al., 2024a] (1–3 seed; the learned-PE row is 3-seed, EVQ/Geo/DAPE rows are seed 42); and a scarce-channel test on MLA [DeepSeek-AI, 2024a,b] with 16 RoPE channels at 432M (3-seed). We additionally report *robustness* experiments on the same MHA setting, and *supporting/exploratory* single- or two-seed evidence (video DiT, LoRA into LLaMA-3-8B, progressive training) that extends scope rather than anchoring primary claims; tier assignment appears in Table 2.

These scope choices are part of the claim, not a retreat from it: EVQ does not replace inference-time scaling, but changes the training-time substrate on which methods such as YaRN act. We also do not claim universal downstream superiority; the target is PE-mechanism isolation at endpoints where positional allocation is isolated and strongly stressed.

Contributions.

- **Finite-spectral-budget view.** We identify RoPE training-time frequency-channel allocation as a third design axis, complementary to operator design and inference-time range scaling; geometric RoPE is an inherited allocation convention, not a consequence of the RoPE operator.
- **Closed-form allocation family.** We introduce EVQ-COSH: a closed-form one-parameter inverse-CDF allocation family with zero learned parameters, in which geometric RoPE is recovered as the $\tau \rightarrow 0$ limit of the pure-tether sub-family. The method modifies only the inverse-frequency initialization.
- **Deployable operating-point rule.** A small- τ softmax-transport argument gives the structural scaling $\tau^* \propto d_{\text{head}} / \sqrt{L}$ with convention-dependent prefactor; $\tau = d_{\text{head}} / \sqrt{L}$ is used as a robust default inside an empirically flat basin rather than a claimed global optimum.
- **Three mechanism stress tests.** Matched-scale EVQ+YaRN tests substrate–range complementarity; PE-dominant extrapolation against DAPE tests whether allocation shape alone reduces extrapolation gaps; compressed-RoPE MLA is the scarce-channel regime where allocation quality is predicted to matter more strongly.

2 Related Work

Positional encodings can be separated along three axes: the *operator* that injects positional information, the *inference-time* mechanism that extends its usable range, and *training-time frequency allocation* / frequency-usage analysis of what channels actually do after training. We position EVQ-Cosh in each axis below.

Axis A: Positional-encoding operators. Transformers rely on positional structure to recover order [Vaswani et al., 2017]; the design space spans absolute encodings, relative biases [Shaw et al., 2018, Raffel et al., 2020], additive linear biases (ALiBi; Press et al., 2022), kernelized biases (KERPLE; Chi et al., 2022), rotary position embedding (RoPE; Su et al., 2024), magnitude-decay rotary (XPOS; Sun et al., 2023), learned positional functions (FIRE, Li et al., 2024; DAPE, Zheng et al., 2024a), and operator variants such as HoPE/hyperbolic rotary encoding [Dai et al., 2025] and PaTH [Yang et al., 2025]. EVQ-Cosh is not a new operator: it preserves the RoPE rotation and uses cosh only as the closed-form solution of a variational ODE on log-frequency density—modifying how the fixed number of frequency channels is distributed along the spectrum. In the PE-dominant diagnostic (§4.3) this zero-learned-parameter allocation change attains lower seed-42 extrapolation PPL

than the DAPE-style learned positional-operator baseline, isolating the frequency-structure effect from operator capacity in that tested protocol.

Axis B: Inference-time and post-training context extension. NTK-aware scaling [Rozière et al., 2023], positional interpolation [Chen et al., 2023], YaRN [Peng et al., 2024], and base-limit analyses [Xu et al., 2024] rescale or bound RoPE positions; in particular, base-limit analyses study how the RoPE base bounds attainable context length, whereas EVQ holds the base fixed and reallocates finite channels within that span. SelfExtend [Jin et al., 2024] uses grouped attention without fine-tuning; LongRoPE [Ding et al., 2024] and PoSE [Zhu et al., 2024] use progressive or skip-wise strategies; LongRoPE2 [Shang et al., 2025] searches per-channel scaling factors by evolution on a frozen checkpoint and combines this with mixed-context-window adaptation; CLEX [Chen et al., 2024] trains a continuous extrapolation module. These methods start from a fixed RoPE allocation and focus on range extension, post-hoc rescaling, or adaptation after that allocation is fixed. EVQ-Cosh is orthogonal: it changes the *training-time* frequency substrate itself, and in our experiments (§4.2) composes with YaRN under matched YaRN scale. *Relation to LongRoPE2.* LongRoPE2 is the closest post-hoc per-channel rescaling method to EVQ, but the two target different stages: LongRoPE2 searches scaling factors for a pretrained checkpoint, whereas EVQ specifies the training-time frequency allocation before optimization begins. The methods are therefore conceptually composable; we leave empirical EVQ+LongRoPE2 composition to future work.

Axis C: Frequency-usage and allocation analysis. Barbero et al. [2025] show empirically that high-frequency channels specialize in positional patterns while low-frequency channels carry semantic content, an asymmetry that geometric allocation does not model. Wang et al. [2024] (Resonance RoPE) identify critical frequencies that should not be perturbed during interpolation, implicitly acknowledging channel inequality but framed around interpolation rather than allocation. Chen et al. [2025] replace specific RoPE components with position-independent ones while retaining high-frequency rotary signals for extrapolation; EVQ instead preserves the full RoPE rotation and reallocates the same finite channel budget by a closed-form training-time density. Wei et al. [2025] propose sequence-length-dependent allocation tailored for video generation; Zhao et al. [2025] correct frequencies at inference time for video diffusion; Qiu et al. [2024] address temporal coherence via noise rescheduling (adjacent, not a positional-encoding method). Concurrent work covers adjacent facets: MHRoPE [Huang et al., 2026] and VLM-HoPE [Li et al., 2025] design or reassign axis-/modality-specific allocations for multimodal RoPE in vision-language models; FoPE [Hua et al., 2025] models each channel as a Fourier series; CARoPE [Veisi et al., 2025] learns dynamic per-head frequencies; Clipped RoPE [Li et al., 2026] soft-clips low frequencies post-hoc. To our knowledge, none derives a closed-form training-time spectral density for standard RoPE from a variational principle.

Evaluation for long-context behavior. Long-context evaluation spans synthetic retrieval diagnostics and broader task suites [Bai et al., 2024b, Hsieh et al., 2024]. Prior work shows that long contexts can fail even when relevant information is nominally present [Liu et al., 2024], and that positional encoding modifications can mitigate such failures [Zhang et al., 2024]. We use both PE-dominant settings (isolating frequency-allocation effects) and retrieval-heavy evaluation (testing systems-level claims), treating incomplete larger-scale evidence conservatively.

Positioning. EVQ-Cosh targets the unfilled cell of a training-time, closed-form, zero-learned-parameter frequency allocation for standard RoPE before optimization begins. It does not compete with range-extension methods by offering another rescaling rule, nor with learned PE methods by adding operator capacity; it changes the pretraining substrate—the frequency table that subsequent RoPE-based range-scaling and operator-adaptation methods inherit. Per-channel learned/searched methods [Shang et al., 2025, Veisi et al., 2025, Hua et al., 2025] target distinct stages or parameterizations, making the families stage-distinct and potentially composable; empirical head-to-head benchmarking is left to future work. A compact method comparison is provided in the appendix; EVQ-Cosh occupies the training-time, closed-form, zero-parameter allocation cell.

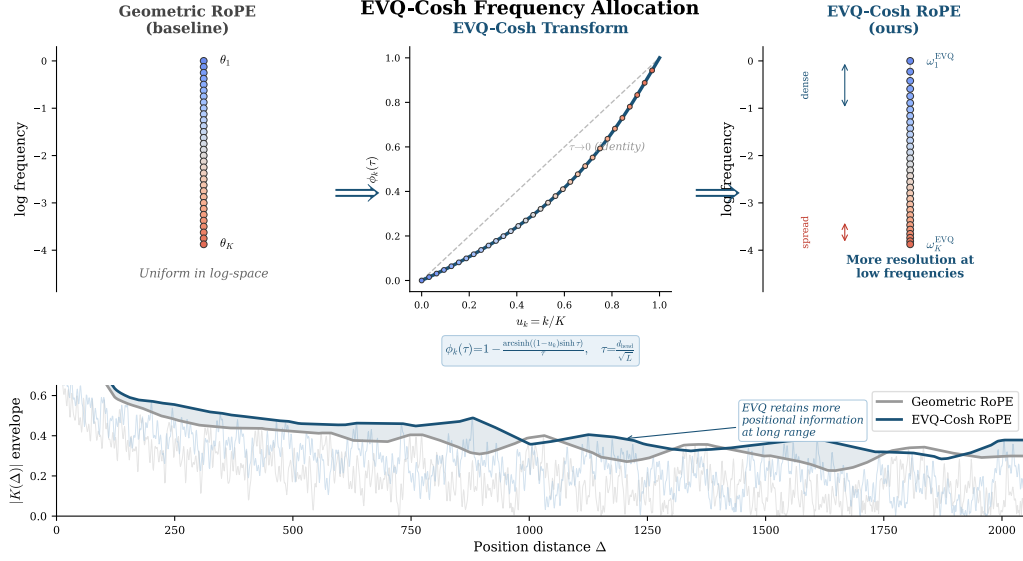


Figure 1: **EVQ-Cosh frequency allocation.** Geometric RoPE places channels uniformly in log-frequency; EVQ-Cosh applies the closed-form inverse-CDF warp to reallocate the same finite set of RoPE channels. The RoPE rotation, architecture, and optimizer are unchanged, and no learned parameters are added. Bottom: exact-kernel phase-collision diagnostic.

Table 1: Epistemic map of the EVQ-Cosh derivation. The variational core (rows 1–2) is exact conditional on the broadband surrogate; the practical recipe (rows 3–4) is a deliberate engineering choice whose operating range is calibrated and validated empirically (§3.7).

Component	Status	Role
Broadband surrogate $\mathcal{C}_{\text{app}}$	Functional approximation of the <i>collision objective</i> (not pointwise approximation of the kernel K)	Yields a tractable variational problem; functionally checked in Appendix A.6
ODE and closed-form family	Exact conditional on $\mathcal{C}_{\text{app}}$	Theoretical core; geometric RoPE is the $\tau \rightarrow 0$ limit of the pure-tether sub-family
Pure-tether branch	Practical branch choice of the stationary ODE	Gives a closed-form invertible quantile warp; forcing branch is CDF/ L^1 -suppressed at typical bases, not pointwise; large τ can amplify warp error
τ^* scaling rule	Semi-analytic operating-point selector (not a global minimizer)	Moves allocation away from the degenerate $\tau \rightarrow 0$ edge into a wide non-geometric basin

137 3 Theory and Method

138 Before any derivation we state the epistemic status of each component (Table 1), so that the varia-
139 tional core is not conflated with the practical recipe.

140 We do not claim an analytic solution to the full trained-transformer objective or direct optimization of
141 the exact oscillatory RoPE kernel. The theorem-level analytic claim is conditional on the broadband
142 surrogate $\mathcal{C}_{\text{app}}$: its Green-kernel relaxation yields a convex variational problem with an elementary
143 inverse-CDF minimizer. The surrogate is validated functionally, not pointwise (Appendix A.6: 24–
144 92% collision-score reduction across 12 configs); downstream PPL/passkey measures trained-model
145 consequences. The pure-tether branch (Theorem 1) is a closed-form design choice, not a claim that
146 Fisher forcing vanishes. The rule $\tau^* = d_{\text{head}} / \sqrt{L}$ (Proposition 1) is an operating-point selector with
147 convention-dependent $\mathcal{O}(1)$ scale, not a global optimum.

3.1 RoPE as frequency allocation

Let $\phi \in [0, 1]$ parameterize log-frequency, $\omega(\phi) = b^{-\phi}$. Let d_{rot} denote the number of dimensions to which RoPE is applied and $K = d_{\text{rot}}/2$ the number of rotary frequency pairs. For standard MHA, $d_{\text{rot}} = d_{\text{head}}$; for compressed-RoPE architectures such as MLA (§4.4), d_{rot} may be smaller than the full attention head dimension d_{head} , in which case the inverse-CDF grid uses $K = d_{\text{rot}}/2$ while the operating-point scale uses the effective attention-head dimension d_{eff} specified for that architecture (we take $d_{\text{eff}} = d_{\text{head}}$ for MLA; see §4.4 and Appendix C.1). We use a positive probability density $\rho \in C^2([0, 1])$ with $\int_0^1 \rho(\phi) d\phi = 1$ to parameterize a family of discrete RoPE allocations via inverse-CDF quantization: $\phi_k = F_\rho^{-1}(u_k)$ with $F_\rho(\phi) = \int_0^\phi \rho(s) ds$ and grid u_k (Figure 1). Geometric RoPE corresponds to $\rho \equiv 1$.

3.2 A surrogate collision functional

Under a smooth, bounded-support distance prior $D \in W^{m,1}(\mathbb{R})$ ($m \geq 1$; the regime relevant to autoregressive language models with finite effective context), the exact phase-collision kernel is $K(\phi_1, \phi_2) = \int D(\Delta) \cos(\omega(\phi_1)\Delta) \cos(\omega(\phi_2)\Delta) d\Delta$; it is oscillatory and is not pointwise approximated by a smooth two-term model. We therefore introduce the following surrogate collision quadratic form as the object of our variational analysis:

$$\mathcal{C}_{\text{app}}[\rho] = \frac{\alpha}{2} \int_0^1 \rho(\phi)^2 d\phi + \frac{\beta}{2} \iint_{[0,1]^2} \rho(\phi)\rho(\psi) \min(\phi, \psi) d\phi d\psi, \quad (1)$$

with constants $\alpha, \beta > 0$ fitted from the exact kernel K on the deployed discrete channel grid ($K \in \{16, 32\}$; Appendix A.11), giving $\alpha \approx 1/d_{\text{rot}}$. The diagonal $\alpha \int \rho^2$ penalizes channel-load concentration. The second term has the identity $\iint \rho(\phi)\rho(\psi) \min(\phi, \psi) d\phi d\psi = \int_0^1 (\int_s^1 \rho(\phi) d\phi)^2 ds$; it penalizes squared cumulative spectral mass shared above every log-frequency threshold. This makes $\min(\phi, \psi)$ —the Green kernel of $-\partial_\phi^2$ on $[0, 1]$ —a tractable cumulative-overlap model, not an asserted pointwise approximation to the oscillatory kernel. $\mathcal{C}_{\text{app}}$ is a discrete-grid surrogate, not the continuum stationary-phase asymptotic of K (which would give a ϕ -dependent diagonal coefficient; see Appendix A.5). We validate $\mathcal{C}_{\text{app}}$ functionally: allocations derived from it reduce the exact-kernel collision score by 24–92% across 12 configurations, and a collision-only oracle search reaches comparable reductions (Appendices A.6, A.14).

3.3 Constrained stationary allocation

Minimizing $\mathcal{C}_{\text{app}}[\rho]$ in (1) subject to $\int_0^1 \rho(\phi) d\phi = 1$ with Lagrange multiplier ν , the first variation gives $\alpha\rho(\phi) + \beta g(\phi) + \nu = 0$ where $g(\phi) = \int_0^1 \rho(\psi) \min(\phi, \psi) d\psi$. Since $g''(\phi) = -\rho(\phi)$, every interior stationary density satisfies the homogeneous equation $\rho''(\phi) - \tau^2 \rho(\phi) = 0$ with $\tau = \sqrt{\beta/\alpha}$. The pre-differentiated stationarity and mass normalization give the derivative boundary conditions $\rho'(0) = -\tau^2$ and $\rho'(1) = 0$ (Appendix A).

Theorem 1 (Exact stationary allocation under the broadband surrogate). *Under (1) with $\alpha > 0, \beta \geq 0$ (so $\mathcal{C}_{\text{app}}$ is convex; Appendix A) and constraints $\rho \in C^2([0, 1])$, $\rho > 0$, $\int_0^1 \rho = 1$, the unique constrained minimizer is*

$$\rho_\tau(\phi) = \frac{\tau \cosh(\tau(1-\phi))}{\sinh \tau}, \quad \tau = \sqrt{\beta/\alpha}, \quad (2)$$

strictly positive on $[0, 1]$ for $\beta > 0$; at $\beta = 0$ the minimizer is the $\tau \rightarrow 0$ limit $\rho_0 \equiv 1$ (uniform/geometric).

Fisher-forcing extension (Appendix). Adding a Fisher utility $-\mu \int \rho b^{-2\phi} d\phi$ to (1) changes the stationary equation to $\rho'' - \tau^2 \rho = \gamma b^{-2\phi}$, whose non-resonant solution includes a particular term $\gamma b^{-2\phi} / ((2 \log b)^2 - \tau^2)$. EVQ-Cosh corresponds to the homogeneous $\gamma = 0$ branch; the forced extension is analyzed as a perturbation in Appendix A.16. Our practical method uses the homogeneous branch because it gives a closed-form invertible CDF; the forced branch is a controlled but nonzero residual, not zero pointwise.

191 3.4 Closed-form EVQ and the geometric limit

192 The pure-tether branch yields an inverse-CDF warp

$$\phi_k(\tau) = 1 - \frac{1}{\tau} \operatorname{arcsinh}((1 - u_k) \sinh \tau), \quad u_k = \frac{k + \frac{1}{2}}{K}, \quad (3)$$

193 and inverse frequencies $b^{-\phi_k(\tau)}$.

194 **Theorem 2** (Closed-form pure-tether allocation and geometric limit). *For $\tau > 0$, the stationary density in Theorem 1 (the homogeneous, pure-tether branch) has CDF*
 195 *$F_\tau(\phi) = 1 - \sinh(\tau(1-\phi))/\sinh \tau$, so inverse-CDF quantization gives $\phi_k(\tau) = 1 -$*
 196 *$\tau^{-1} \operatorname{arcsinh}((1-u_k) \sinh \tau)$. As $\tau \rightarrow 0$, $\phi_k(\tau) \rightarrow u_k$ and geometric RoPE is recovered within*
 197 *the pure-tether sub-family; the Fisher-forced extension (Appendix A.2) does not vanish in this limit.*

199 3.5 Practical EVQ instantiation

200 For an effective attention-head dimension d_{eff} and training length L , the complete implementation
 201 is

$$\tau = \frac{d_{\text{eff}}}{\sqrt{L}}, \quad u_k = \frac{k + \frac{1}{2}}{K}, \quad \omega_k^{\text{EVQ}} = b^{-\phi_k(\tau)}, \quad K = \frac{d_{\text{rot}}}{2}. \quad (4)$$

202 For standard MHA, $d_{\text{eff}}=d_{\text{rot}}=d_{\text{head}}$; for compressed-RoPE MLA, $K=d_{\text{rot}}/2$ counts only the ro-
 203 tary channels while d_{eff} follows the architecture-specific convention (we take $d_{\text{eff}}=d_{\text{head}}$; see §4.4
 204 and Appendix C.1). The construction is deliberately minimal: set τ , compute midpoint quan-
 205 tiles, evaluate the warp, replace inverse frequencies. No auxiliary loss, no extra parameters, no
 206 optimizer dependence. The endpoint grid $u_k=k/K$ is useful as an analytic anchored variant be-
 207 cause $u_0=0 \Rightarrow \phi_0=0$ (preserves $\omega_0=1$); our deployed implementation uses midpoint quantization in
 208 Eq. (4), which trades exact anchoring for avoiding endpoint cells. We do not rely on $\omega_0=1$ in any
 209 trained model, and the endpoint/midpoint choice shifts PPL by $<1\%$ across all $K \geq 16$ settings we
 210 tested, well inside the flat empirical basin.

211 EVQ-Cosh changes only the inverse-frequency initialization—distinct from learnable PE ap-
 212 proaches that rely on in-distribution training dynamics to discover the allocation through extra pa-
 213 rameters. The full recipe is Equation (4) plus $\tau=d_{\text{eff}}/\sqrt{L_{\text{train}}}$: zero learned parameters, architecture
 214 unchanged.

215 3.6 Waterbed inequality

216 Under the broadband surrogate, EVQ obeys a waterbed inequality (a corollary of the sharper enve-
 217 lope in Proposition 3, Appendix A.8). Writing P_ρ for the CDF-induced measure and U for the uni-
 218 form distribution on $[0, 1]$, and letting $D_{\text{Burg}}(\rho) := -\int_0^1 \log \rho(\phi) d\phi$ denote the Burg entropy deficit
 219 relative to uniform, the Pearson-divergence companion functional $\mathcal{W}(\rho) = \alpha \chi^2(U \| P_\rho) + \beta \int M_\rho^2$
 220 (related to but distinct from $\mathcal{C}_{\text{app}}$: both vanish at $\rho \equiv 1$ with leading-order $\mathcal{O}(\tau^4)$ growth; they diverge
 221 at large τ) satisfies $\mathcal{W}(\rho) \geq \alpha(e^{D_{\text{Burg}}(\rho)} - 1)$ with equality iff $\rho \equiv 1$. Since $\mathcal{W}(\rho_\tau) = \mathcal{O}(\tau^4)$ and
 222 Proposition 1 (with the $L \rightarrow L_{\text{eff}}^J$ substitution from Appendix A.15) gives $\tau_*^2 \sim d_{\text{head}}^2 / L_{\text{eff}}^J$, EVQ’s
 223 in-range Pearson deviation scales as $\mathcal{W}(\rho_{\tau_*}) = \mathcal{O}(d_{\text{head}}^4 / (L_{\text{eff}}^J)^2)$. Long-range improvement is thus
 224 paid for by a bounded, quartic-in- τ^* in-range cost.

225 3.7 The τ^* scaling law: epistemic stratification

226 **Shape and scale come from separate layers.** The cosh shape is the pre-softmax stationary solu-
 227 tion under $\mathcal{C}_{\text{app}}$ (§3.3–§3.5); the scale τ^* is selected *post*-softmax by the proposition below, with
 228 $U(\tau, L)$ the per-channel post-softmax KL gain from an $\mathcal{O}(\tau^2)$ RoPE logit perturbation (formal def-
 229 inition Appendix A.12). The $1/L$ factor is the softmax Jacobian eigenvalue at the diffuse baseline
 230 $p_0=1/L$ —an algebraic property, not a CLT claim; convention $\frac{1}{2}$ factors from the KL Taylor expan-
 231 sion are absorbed into λ (Appendix A.18).

232 **Proposition 1** (Semi-analytic operating-point scaling under a diffuse baseline). *On the pure-tether*
 233 *family ρ_τ , and assuming a diffuse post-softmax baseline $p_0=1/L$, the Pearson- χ^2 channel-load*

234 *stiffness and post-softmax transport utility admit small- τ expansions*

$$S_{\chi^2}(\tau) = c_S \frac{\tau^4}{d_{\text{head}}} + O(\tau^6), \quad c_S = \frac{1}{45}, \quad U(\tau, L) = \frac{d_{\text{head}}}{L} [Q_0 + \tau^2 Q_1(L, b) + O(\tau^4)],$$

235 *with $Q_1(L, b) = \int_0^1 \eta(\phi) q(Lb^{-\phi}) d\phi$, $\eta(\phi) = (1-\phi)^2/2 - 1/6$, and $q(x) = 1/2 + \sin(2x)/(4x) -$*
 236 *$(\sin x/x)^2$ (Appendix A.12). The leading-order stationary point of the truncated objective $\mathcal{F}(\tau) =$*
 237 *$\frac{1}{2} S_{\chi^2}(\tau) - \lambda U(\tau, L)$ therefore satisfies*

$$\tau_*^2 = 45 \lambda Q_1(L, b) \cdot \frac{d_{\text{head}}^2}{L}, \quad \tau_* \propto \frac{d_{\text{head}}}{\sqrt{L}}.$$

238 *The d_{head} factor and $L^{-1/2}$ exponent follow from the combined small- τ scalings; the rule balances*
 239 *an $\mathcal{O}(\tau^2)$ utility gain (RoPE logit perturbation transported through a diffuse softmax with $1/L$*
 240 *Jacobian curvature) against an $\mathcal{O}(\tau^4)$ channel-load stiffness. Real τ_* requires $Q_1(L, b) > 0$, verified*
 241 *across the tested grid ($Q_1 \in [0.008, 0.032]$; Appendix A.10). The remaining $\mathcal{O}(1)$ prefactor depends*
 242 *on normalization convention and is fixed empirically inside a flat basin, not claimed as a global*
 243 *optimum.*

244 **Three epistemic tiers.** The small- τ expansion in Proposition 1 *derives the structural d_{head} factor*
 245 *and $L^{-1/2}$ exponent; at practical τ (e.g., $\tau=4$ in the PE-dominant setting) the exact cosh family is*
 246 *used directly, not the Taylor truncation, and the deployed $L^{-0.500}$ rule lies within $< 1\%$ PPL of the*
 247 *best observed sweep points. The Pearson χ^2 stiffness $S_{\chi^2}(\rho) = \text{Var}_{\rho d\phi}(1/\rho)$ is the canonical load-*
 248 *variance penalty under inverse-CDF cell-width $1/\rho(\phi)$ (Appendix A.17); the Lagrange multiplier*
 249 *$\lambda=1$ is a unit convention, with an exact-kernel sweep giving $c_{\text{coll}}=1.171$ vs. surrogate $\sqrt{45}Q_1 \approx 1.19$*
 250 *(1.6% relative, CV 0.28%, 9 configs; Table 6). For trained, non-diffuse attention the same expansion*
 251 *replaces $1/L$ by a softmax-Jacobian Rayleigh quotient $1/L_{\text{eff}}^J$ (Appendix A.15); we register direct*
 252 *measurement at $L_{\text{train}} \geq 16\text{K}$ as a falsifiable test of the diffuse-baseline approximation.*

253 **Advance prediction versus calibration.** Before training, the theory fixes the conditional cosh
 254 shape and predicts that useful operating points should scale with d_{head} and approximately $L^{-1/2}$
 255 under the diffuse-softmax assumptions. It does *not* predict the unit prefactor, prove Pearson χ^2 is
 256 unique, identify d_{eff} for compressed-RoPE architectures, or establish $L_{\text{eff}}^J = L$ in trained attention.
 257 Those pieces are respectively a unit convention checked against the observed basin, a modeling
 258 choice, an architecture-specific calibration, and an unmeasured falsification target. Thus the de-
 259 ployed rule is a basin selector motivated by structural theory, not a parameter-free prediction of the
 260 empirical optimum.

261 **Basin, not point estimate.** With $\lambda=1$, the formula falls within $< 1\%$ PPL of the best observed
 262 sweep points across 27 configurations; pure geometric ($\tau=0$) is the natural negative control and
 263 falls 10–46% outside (Appendices A.11, A.10). The bare formula stays inside this basin for our
 264 primary $b \geq 500\text{K}$ regime; at small b and large L the explicit $c_{\text{pred}}(L, b) = \sqrt{45 Q_1(L, b)}$ form is
 265 recommended. EVQ and YaRN correct orthogonal deficiencies (shape vs. range; Appendix Fig. 3),
 266 motivating the EVQ+YaRN composition test in §4.

267 4 Experiments

268 4.1 Setup and evidence tiering

269 **Evaluation scope.** This is a positional-encoding mechanism study. PE-diagnostic endpoints (ex-
 270 trapolation PPL, passkey, NIAH/RULER-style retrieval, gold-answer NLL) are our primary mech-
 271 anistic endpoints; PK denotes teacher-forced NLL-gap retrieval unless explicitly marked as au-
 272 toregressive exact match. Downstream accuracy at these scales is a non-regression check rather
 273 than a primary claim. EVQ-Cosh modifies only the training-time allocation and adds no learned
 274 parameters—it is neither an inference-time rescaler (YaRN/NTK/LongRoPE) nor a learned posi-
 275 tional operator (DAPE/FIRE).

276 Experiments test three questions: (Q1) does EVQ improve over Geo at fixed training length?
 277 (Q2) does EVQ compose with inference-time scaling under matched YaRN scale? (Q3) does the

278 τ^* rule land in the best observed sweep basin? Core runs train from scratch with fixed architecture,
 279 changing only the RoPE inverse-frequency initialization. We use $b=500K$, predicted $\tau^*(L)$, and
 280 GPT-style decoder stacks [Radford et al., 2019] with $d_{\text{head}}=64$ across 50M–750M. Primary text
 281 data are FineWeb-Edu [Penedo et al., 2024] and controlled passkey-mix settings; TinyStories [El-
 282 dan and Li, 2023] appears only in supporting quick-validation runs. Full hyperparameters are in
 283 Appendix D.

284 Evidence is tiered in Table 2. Primary claims use EVQ×YaRN (3-seed), PE-dominant extrapolation
 285 (seed-42 diagnostic for Geo/DAPE/EVQ; learned-PE row 3-seed), and MLA (3-seed). Robustness
 286 rows are 3-seed; supporting/exploratory rows are 1–2 seed. No abstract or introduction claim rests
 287 on tier-3 evidence.

Table 2: Evidence tier by setting. Primary systems/MLA claims are 3-seed; the PE-dominant Geo/DAPE/EVQ contrast is seed 42.

Tier	Setting	Seeds	Location
Primary I	EVQ+YaRN, 454M MHA, $L_{\text{train}}=2048$, 4/6/8× extrapolation	3	§4.2
Primary II	PE-dominant 128→8K (DAPE; learned-PE row 3-seed, others seed 42)	seed 42 / 3	§4.3
Primary III	MLA 432M, 16 RoPE channels, 500M tokens	3	§4.4
Robustness	FineWeb-Edu consistency, base sweep, capability preservation	3	§4.5
Supporting	Multi-scale raw-PPL one-seed rows	1	§4.5, App. D
Supporting	Video DiT 129M / 382M	1–2	§4.5, App. D
Supporting	LoRA into LLaMA-3-8B-Instruct	1	§4.5, App. D
Supporting	Progressive training	1	§4.5, App. D
Supporting	MLA 1B-token schedule-sensitivity check	1	§4.4, App. D

288 4.2 Primary I: EVQ×YaRN matched-scale substrate test

289 *Substrate vs. range stress test.* If EVQ only changed effective range, the same YaRN scale should
 290 largely erase its advantage; if it changes the training-time substrate, YaRN should have higher
 291 leverage on EVQ than Geo (Table 3, Fig. 2). In the 454M 10% passkey-mix setting (3 seeds),
 292 EVQ+YaRN reaches 100% teacher-forced NLL-gap passkey retrieval at 8K while Geo+YaRN re-
 293 mains at 61%, with the advantage persisting at 12/16K (79%/68% vs. 59%/51%); PPL drops from
 294 82.9 to 70.9 at 8K and from 157.7 to 107.5 at 16K. Appendix Table 20 quantifies the interaction; we
 295 read this as increased matched-scale YaRN leverage, not dominance over every tuned-scale YaRN
 296 baseline.

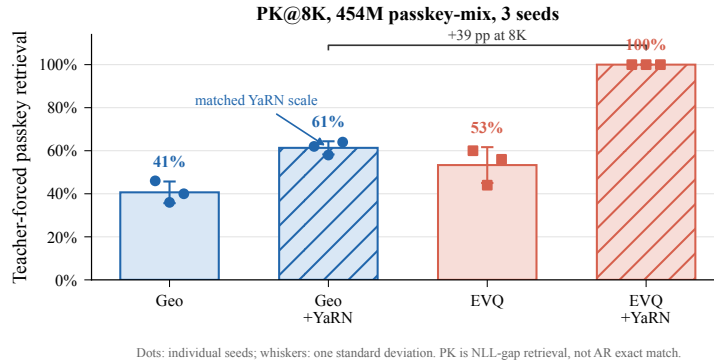


Figure 2: Primary EVQ×YaRN retrieval anchor. Bars show teacher-forced NLL-gap retrieval at 8K (PK) across three seeds; dots are seeds and whiskers are one standard deviation. EVQ+YaRN reaches 100% while Geo+YaRN remains at 61%.

Table 3: Training-time and inference-time positional optimization (454M, $L_{\text{train}}=2048$, 10% passkey mix). YaRN uses the same fixed scale $s=8$ on Geo and EVQ. NLL-gap retrieval (PK) at 8K is 3-seed mean $\pm$ std; other NLL-gap/PPL entries are 3-seed means.

Method	Type	Seeds	NLL-gap@8K	@12K	@16K	PPL@8K / 16K
Geo	baseline	3	41 $\pm$ 5%	57%	51%	161.9 / 253.2
Geo+YaRN	infer	3	61 $\pm$ 3%	59%	51%	82.9 / 157.7
EVQ	train	3	53 $\pm$ 8%	63%	50%	150.3 / 229.5
EVQ+YaRN	train+infer	3	100$\pm$0%	79%	68%	70.9 / 107.5

4.3 Primary II: PE-dominant allocation stress test

Allocation-shape stress test. In a DAPE-style 128 $\rightarrow$ 8K protocol [Zheng et al., 2024a], we ask whether allocation shape alone can reduce a PE-dominant extrapolation gap without adding a learned positional operator. Table 4: in the retained seed-42 Geo/DAPE/EVQ contrast, EVQ reaches 333.7 PPL (vs. Geo 513.7, DAPE 455.3) with no learned parameters. We read this as competitiveness in this tested protocol, not comprehensive dominance over DAPE. Complementary $L_{\text{train}}=256$ sweeps support the predicted $\tau^*=4.0$ setting; Appendix Fig. 10 and Table 23 show these supporting curves.

Table 4: PE-dominant summary (125M, $L_{\text{train}}=128$, FineWeb-Edu). Geo, DAPE, and EVQ are seed 42; Learnable τ is 3-seed (42/137/256), mean $\pm$ std.

Method (128 $\rightarrow$ 8K)	Extra params	PPL@128	PPL@8K	Δ vs Geo
Geo	0	184.9	513.7	–
Learnable τ	1	181.2 $\pm$ 1.3	437.9 $\pm$ 12.2	–14.8%
DAPE	32	183.6	455.3	–11.4%
EVQ	0	182.0	333.7	–35.0%

4.4 Primary III: scarce-channel stress test (MLA)

Scarce-channel stress test. MLA [DeepSeek-AI, 2024a,b] compresses RoPE to $d_{\text{rope}}=32$ (16 channels) vs. 64 in MHA at $d_{\text{head}}=128$, where the finite-budget view predicts stronger EVQ effects. For MLA, d_{rot} determines quantized rotary channels, while $d_{\text{eff}}=d_{\text{head}}$ is the calibrated transport-scale convention used in this stress test. On a 432M MLA transformer trained for 500M tokens at $L_{\text{train}}=8192$ (3 seeds), EVQ reduces $2\times$ extrapolation PPL from 138.8 to 95.6 (–31.1%) with +1.1% in-distribution cost, exceeding Geo+YaRN at matched scale $s=4$ (117.9; Appendix C.1). This is larger than the matched MHA improvement (–13.3%; Table 21), consistent with scarce channels amplifying allocation effects. The result supports allocation sensitivity at the stated convention; it does not identify the optimal MLA d_{eff} , for which the direct $d_{\text{rope}}/\sqrt{L}$ ablation remains required.

4.5 Robustness and supporting evidence

Multi-scale raw extrapolation (Table 21) keeps in-range PPL within $\pm 1.7\%$ while long-range PPL improves 10–46%; capability checks (Table 25) confirm non-destructiveness. Supporting (1–2 seed) evidence covers video DiT, LoRA into LLaMA-3-8B, progressive training, and a single-seed MLA saturation check; full breakdowns are in Appendix D.

5 Discussion and Limitations

This mechanism study isolates training-time allocation and claims PE diagnostics, not universal accuracy. Theory is exact only for the surrogate: realistic priors and L_{eff}^J are unmeasured; τ is a basin default; MLA/per-head choices remain unablated. Composition is YaRN-specific: EVQ4+NTK loses to Geo+NTK at $L=256$, $32\times$ (331.4 vs. 198.1; Appendix Table 20). Video/LoRA/progressive rows are supporting. EVQ adds no inference operations but reduces neither attention/KV-cache complexity nor training FLOPs; energy and deployment gains require direct validation.

References

- Yushi Bai, Xin Lv, Jiajie Zhang, Yuze He, Ji Qi, Lei Hou, Jie Tang, Yuxiao Dong, and Juanzi Li. LongAlign: A recipe for long context alignment of large language models. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 1376–1395, 2024a. doi: 10.18653/v1/2024.findings-emnlp.74.
- Yushi Bai, Xin Lv, Jiajie Zhang, Hongchang Lyu, Jiankai Tang, Zhidian Huang, Zhengxiao Du, Xiao Liu, Aohan Zeng, Lei Hou, Yuxiao Dong, Jie Tang, and Juanzi Li. LongBench: A bilingual, multitask benchmark for long context understanding. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*, 2024b.
- Federico Barbero, Alex Vitvitskyi, Christos Perivolaropoulos, Razvan Pascanu, and Petar Veličković. Round and round we go! what makes rotary positional encodings useful? In *International Conference on Learning Representations*, 2025.
- Guanzheng Chen, Xin Li, Zaiqiao Meng, Shangsong Liang, and Lidong Bing. CLEX: Continuous length extrapolation for large language models. In *International Conference on Learning Representations*, 2024.
- Shouyuan Chen, Sherman Wong, Liangjian Chen, and Yuandong Tian. Extending context window of large language models via positional interpolation. *arXiv preprint arXiv:2306.15595*, 2023.
- Yuhan Chen, Ang Lv, Jian Luan, Bin Wang, and Wei Liu. HoPE: A novel positional encoding without long-term decay for enhanced context awareness and extrapolation. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 23044–23056, 2025. doi: 10.18653/v1/2025.acl-long.1123.
- Ta-Chung Chi, Ting-Han Fan, Peter J. Ramadge, and Alexander I. Rudnicky. KERPLE: Kernelized relative positional embedding for length extrapolation. In *Advances in Neural Information Processing Systems*, volume 35, 2022.
- Chang Dai, Hongyu Shan, Mingyang Song, and Di Liang. HoPE: Hyperbolic rotary positional encoding for stable long-range dependency modeling in large language models. *arXiv preprint arXiv:2509.05218*, 2025.
- DeepSeek-AI. DeepSeek-V2: A strong, economical, and efficient mixture-of-experts language model. *arXiv preprint arXiv:2405.04434*, 2024a.
- DeepSeek-AI. DeepSeek-V3 technical report. *arXiv preprint arXiv:2412.19437*, 2024b.
- Yiran Ding, Li Lyna Zhang, Chengruidong Zhang, Yuanyuan Xu, Ning Shang, Jiahang Xu, Fan Yang, and Mao Yang. LongRoPE: Extending LLM context window beyond 2 million tokens. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *PMLR*, 2024.
- Ronen Eldan and Yuanzhi Li. TinyStories: How small can language models be and still speak coherent english? *arXiv preprint arXiv:2305.07759*, 2023.
- Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, et al. The Llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024.
- Cheng-Ping Hsieh, Simeng Sun, Samuel Kriman, Shantanu Acharya, Dima Rekish, Fei Jia, Yang Zhang, and Boris Ginsburg. RULER: What’s the real context size of your long-context language models? *arXiv preprint arXiv:2404.06654*, 2024.
- Ermo Hua et al. Fourier position embedding: Enhancing attention’s periodic extension for length generalization. In *Proceedings of the 42nd International Conference on Machine Learning*, 2025. *arXiv:2412.17739*.
- Jie Huang, Xuejing Liu, Sibong Song, Ruibing Hou, Hong Chang, Junyang Lin, and Shuai Bai. Revisiting multimodal positional encoding in vision–language models. In *International Conference on Learning Representations*, 2026. *arXiv:2510.23095*.
- Hongye Jin, Xiaotian Han, Jingfeng Yang, Zhimeng Jiang, Zirui Liu, Chia-Yuan Chang, Huiyuan Chen, and Xia Hu. LLM maybe LongLM: SelfExtend LLM context window without tuning. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *PMLR*, pages 22099–22114, 2024.

378 Weijie Kong, Qi Tian, Zijian Zhang, Rox Min, Zuozhuo Dai, Jin Zhou, Jiangfeng Xiong, Xin Li,
379 Bo Wu, et al. HunyuanVideo: A systematic framework for large video generative models. *arXiv*
380 *preprint arXiv:2412.03603*, 2024.

381 Haoran Li, Sucheng Ren, Alan Yuille, and Feng Wang. CoPE: Clipped RoPE as a scalable free
382 lunch for long context LLMs. *arXiv preprint arXiv:2602.05258*, 2026.

383 Haoran Li et al. HoPE: Hybrid of position embedding for long context vision–language models. In
384 *Advances in Neural Information Processing Systems*, 2025. arXiv:2505.20444.

385 Shanda Li, Chong You, Guru Guruganesh, Joshua Ainslie, Santiago Ontanon, Manzil Zaheer, Sumit
386 Sanghai, Yiming Yang, Sanjiv Kumar, and Srinadh Bhojanapalli. Functional interpolation for
387 relative positions improves long context transformers. In *International Conference on Learning*
388 *Representations*, 2024.

389 Nelson F. Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and
390 Percy Liang. Lost in the middle: How language models use long contexts. *Transactions of the*
391 *Association for Computational Linguistics*, 12:157–173, 2024.

392 Xin Ma, Yaohui Wang, Gengyun Jia, Xinyuan Chen, Ziwei Liu, Yuan-Fang Li, Cunjian Chen,
393 and Yu Qiao. Latte: Latent diffusion transformer for video generation. *arXiv preprint*
394 *arXiv:2401.03048*, 2024.

395 Richard Yuanzhe Pang, Alicia Parrish, Nitish Joshi, Nikita Nangia, Jason Phang, Angelica Chen,
396 Vishakh Padmakumar, Johnny Ma, Jana Thompson, He He, and Samuel Bowman. QuALITY:
397 Question answering with long input texts, yes! In *Proceedings of the 2022 Conference of the*
398 *North American Chapter of the Association for Computational Linguistics: Human Language*
399 *Technologies*, pages 5336–5358, 2022. doi: 10.18653/v1/2022.naacl-main.391.

400 Guilherme Penedo, Hynek Kydlíček, Loubna Ben Allal, Anton Lozhkov, Margaret Mitchell, Colin
401 Raffel, Leandro von Werra, and Thomas Wolf. The FineWeb datasets: Decanting the web for the
402 finest text data at scale. *arXiv preprint arXiv:2406.17557*, 2024.

403 Bowen Peng, Jeffrey Quesnelle, Honglu Fan, and Enrico Shippole. YaRN: Efficient context window
404 extension of large language models. In *International Conference on Learning Representations*,
405 2024.

406 Ofir Press, Noah A. Smith, and Mike Lewis. Train short, test long: Attention with linear biases
407 enables input length extrapolation. In *International Conference on Learning Representations*,
408 2022.

409 Haonan Qiu, Menghan Xia, Yong Zhang, Yingqing He, Xintao Wang, Ying Shan, and Ziwei Liu.
410 FreeNoise: Tuning-free longer video diffusion via noise rescheduling. In *International Confer-*
411 *ence on Learning Representations*, 2024.

412 Alec Radford, Jeffrey Wu, Rewon Child, David Luan, Dario Amodei, and Ilya Sutskever. Language
413 models are unsupervised multitask learners. *OpenAI Technical Report*, 2019.

414 Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi
415 Zhou, Wei Li, and Peter J. Liu. Exploring the limits of transfer learning with a unified text-to-text
416 transformer. *Journal of Machine Learning Research*, 21(140):1–67, 2020.

417 Baptiste Rozière, Jonas Gehring, Fabian Gloeckle, Sten Sootla, Itai Gat, Xiaoqing Ellen Tan, Yossi
418 Adi, Jingyu Liu, Romain Sauvestre, Tal Remez, et al. Code llama: Open foundation models for
419 code. *arXiv preprint arXiv:2308.12950*, 2023.

420 Ning Shang, Li Lyna Zhang, Siyuan Wang, Gaokai Zhang, Gilsinia Lopez, Fan Yang, Weizhu Chen,
421 and Mao Yang. LongRoPE2: Near-lossless LLM context window scaling. In *Proceedings of the*
422 *42nd International Conference on Machine Learning*, volume 267 of *PMLR*, 2025.

423 Peter Shaw, Jakob Uszkoreit, and Ashish Vaswani. Self-attention with relative position representa-
424 tions. In *Proceedings of the 2018 Conference of the North American Chapter of the Association*
425 *for Computational Linguistics: Human Language Technologies*, 2018.

426 Nitish Srivastava, Elman Mansimov, and Ruslan Salakhutdinov. Unsupervised learning of video
427 representations using LSTMs. In *Proceedings of the 32nd International Conference on Machine*
428 *Learning*, volume 37 of *PMLR*, pages 843–852, 2015.

429 Jianlin Su, Murtadha Ahmed, Yu Lu, Shengfeng Pan, Bo Wen, and Yunfeng Liu. RoFormer: En-
430 hanced transformer with rotary position embedding. *Neurocomputing*, 568:127063, 2024.

431 Yutao Sun, Li Dong, Barun Patra, Shuming Ma, Shaohan Huang, Alon Benhaim, Vishrav Chaudhary,
432 Xia Song, and Furu Wei. A length-extrapolatable transformer. In *Proceedings of the 61st Annual*
433 *Meeting of the Association for Computational Linguistics*, 2023.

434 Hugo Touvron, Louis Martin, Kevin Stone, Peter Albert, Amjad Almahairi, Yasmine Babaei, Niko-
435 lay Bashlykov, Soumya Batra, Prajjwal Bhargava, Shruti Bhosale, et al. Llama 2: Open founda-
436 tion and fine-tuned chat models. *arXiv preprint arXiv:2307.09288*, 2023.

437 Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez,
438 Lukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Infor-*
439 *mation Processing Systems*, volume 30, 2017.

440 Ali Veisi, Delaram Fartoot, and Hamidreza Amirzadeh. Context-aware rotary position embedding.
441 *arXiv preprint arXiv:2507.23083*, 2025.

442 Wan Team. Wan: Open and advanced large-scale video generative models. *arXiv preprint*
443 *arXiv:2503.20314*, 2025.

444 Suyuchen Wang, Ivan Kobyzev, Peng Lu, Mehdi Rezagholizadeh, and Bang Liu. Resonance RoPE:
445 Improving context length generalization of large language models. In *Findings of the Association*
446 *for Computational Linguistics: ACL 2024*, 2024.

447 Xilin Wei, Xiaoran Liu, Yuhang Zang, Xiaoyi Dong, Pan Zhang, Yuhang Cao, Jian Tong, Haodong
448 Duan, Qipeng Guo, Jiaqi Wang, Xipeng Qiu, and Dahua Lin. VideoRoPE: What makes for
449 good video rotary position embedding? In *Proceedings of the 42nd International Conference on*
450 *Machine Learning*, volume 267 of *PMLR*, 2025.

451 Mingyu Xu, Xin Men, Bingning Wang, Qingyu Zhang, Hongyu Lin, Yaojie Lu, Xianpei Han, and
452 Weipeng Chen. Base of RoPE bounds context length. In *Advances in Neural Information Pro-*
453 *cessing Systems*, volume 37, 2024. doi: 10.52202/079017-2773.

454 An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chengyuan Li, Day-
455 iheng Liu, Fei Huang, et al. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*, 2024a.

456 Songlin Yang, Yikang Shen, Kaiyue Wen, Shawn Tan, Mayank Mishra, Liliang Ren, Rameswar
457 Panda, and Yoon Kim. PaTH attention: Position encoding via accumulating householder transfor-
458 mations. In *Advances in Neural Information Processing Systems*, 2025. NeurIPS 2025 poster.

459 Zhuoyi Yang, Jiayan Teng, Wendi Zheng, Ming Ding, Shiyu Huang, Jiazheng Xu, Yuanming Yang,
460 Wenyi Hong, Xiaohan Zhang, Guanyu Feng, et al. CogVideoX: Text-to-video diffusion models
461 with an expert transformer. *arXiv preprint arXiv:2408.06072*, 2024b.

462 Zhenyu Zhang, Runjin Chen, Shiwei Liu, Zhewei Yao, Olatunji Ruwase, Beidi Chen, Xiaoxia Wu,
463 and Zhangyang Wang. Found in the middle: How language models use long contexts better
464 via plug-and-play positional encoding. In *Advances in Neural Information Processing Systems*,
465 volume 37, 2024.

466 Min Zhao, Guande He, Yixiao Chen, Hongzhou Zhu, Chongxuan Li, and Jun Zhu. RIFLEx: A
467 free lunch for length extrapolation in video diffusion transformers. In *Proceedings of the 42nd*
468 *International Conference on Machine Learning*, volume 267 of *PMLR*, 2025.

469 Chuanyang Zheng, Yihang Gao, Han Shi, Minbin Huang, Jingyao Li, Jing Xiong, Xiaozhe Ren,
470 Michael K. Ng, Xin Jiang, Zhenguo Li, and Yu Li. DAPE: Data-adaptive positional encoding for
471 length extrapolation. In *Advances in Neural Information Processing Systems*, volume 37, 2024a.

472 Zangwei Zheng, Xiangyu Peng, Tianji Yang, Chenhui Shen, Shenggui Li, Hongxin Liu, Yukun
473 Zhou, Tianyi Li, and Yang You. Open-Sora: Democratizing efficient video production for all.
474 *arXiv preprint arXiv:2412.20404*, 2024b.

475 Dawei Zhu, Nan Yang, Liang Wang, Yifan Song, Wenhao Wu, Furu Wei, and Sujian Li. PoSE:
476 Efficient context window extension of LLMs via positional skip-wise training. In *International*
477 *Conference on Learning Representations*, 2024.

A Proof Details

A.1 Collision-only surrogate stationary density

This subsection proves the main-text Theorem 1 (Surrogate stationary allocation) directly from the collision-only surrogate functional; the forced Fisher extension is treated separately in §A.2. Starting from the formal surrogate kernel

$$K_{\text{app}}(\phi, \psi) = \alpha \delta(\phi - \psi) + \beta \min(\phi, \psi), \quad (5)$$

the main-text collision functional is (1), which under the mass normalization $\int_0^1 \rho = 1$ with Lagrange multiplier ν has constrained first variation

$$\alpha \rho(\phi) + \beta g(\phi) + \nu = 0, \quad g(\phi) = \int_0^1 \rho(\psi) \min(\phi, \psi) d\psi. \quad (6)$$

Existence. For rigor, view $\mathcal{C}_{\text{app}}$ first as a functional on $L^2([0, 1])$ and minimize over $\mathcal{A} = \{\rho \in L^2([0, 1]) : \rho \geq 0 \text{ a.e., } \int_0^1 \rho = 1\}$. The $\alpha \|\rho\|_2^2$ term is coercive and weakly lower semicontinuous, the Green-kernel term is positive semidefinite and continuous, and the mass/nonnegativity constraints are weakly closed. A minimizer therefore exists by the direct method. Strict convexity gives uniqueness. The Euler solution derived below is strictly positive, so the nonnegativity constraint is inactive a posteriori and the classical C^2 derivation is justified.

Direct differentiation of g gives

$$g'(\phi) = \int_\phi^1 \rho(\psi) d\psi, \quad g''(\phi) = -\rho(\phi). \quad (7)$$

Differentiating the first variation twice in ϕ eliminates ν (a constant) and yields the homogeneous ODE

$$\rho''(\phi) - \tau^2 \rho(\phi) = 0, \quad \tau = \sqrt{\beta/\alpha}. \quad (8)$$

Evaluating the pre-differentiated stationarity together with $g(0)=0$, $g'(1)=0$, and $\int_0^1 \rho = 1$ gives the derivative boundary conditions

$$\rho'(0) = -\tau^2, \quad \rho'(1) = 0, \quad (9)$$

since differentiating $\alpha \rho(\phi) + \beta g(\phi) + \nu = 0$ once and evaluating at $\phi=1$ gives $\alpha \rho'(1) + \beta g'(1) = 0$ hence $\rho'(1)=0$, and at $\phi=0$ gives $\alpha \rho'(0) + \beta g'(0) = 0$ with $g'(0) = \int_0^1 \rho = 1$, so $\rho'(0) = -\beta/\alpha = -\tau^2$. Substituting $\rho(\phi) = C_1 \cosh(\tau\phi) + C_2 \sinh(\tau\phi)$ into (8) and (9) together with $\int_0^1 \rho = 1$ gives the unique positive interior stationary density

$$\rho_\tau(\phi) = \frac{\tau \cosh(\tau(1-\phi))}{\sinh \tau}, \quad (10)$$

which is main-text Theorem 1. The mass constraint $\int \rho=1$ is not lost: it enters the boundary condition at $\phi=0$ via $g'(0)=\int_0^1 \rho=1$. Positivity holds on the closed interval since $\rho_\tau(\phi) \geq \rho_\tau(1) = \tau / \sinh \tau > 0$ for $\tau > 0$ (with $\rho_\tau(0) = \tau \coth \tau > 0$); the inequality constraint $\rho > 0$ is therefore inactive throughout $[0, 1]$ and the unconstrained Euler–Lagrange solution coincides with the KKT-constrained solution. For the degenerate case $\beta=0$ the ODE reduces to $\rho''=0$ with $\rho'(0)=\rho'(1)=0$ and $\int \rho=1$, forcing $\rho \equiv 1$, which is also the $\tau \rightarrow 0$ limit of (10).

Convexity / PSD of $\mathcal{C}_{\text{app}}$. For $\alpha > 0$ and $\beta \geq 0$, the surrogate is convex because both quadratic terms are positive semidefinite. The diagonal $\alpha \int \rho^2$ is manifestly PSD on L^2 , and the Green-kernel cross term satisfies the identity

$$\iint_{[0,1]^2} f(\phi) f(\psi) \min(\phi, \psi) d\phi d\psi = \int_0^1 \left(\int_s^1 f(u) du \right)^2 ds \geq 0. \quad (11)$$

Thus the stationary density obtained above is the unique constrained *minimizer* (not merely a critical point), and the surrogate is well-defined as a quadratic functional on $L^2([0, 1]) \supset C^2([0, 1])$ with no further regularity assumption needed for the δ -component. Conditional on $\mathcal{C}_{\text{app}}$, this derivation is exact.

513 A.2 Forced Fisher extension

514 Adding a Fisher utility $-\mu \int \rho(\phi) b^{-2\phi} d\phi$ to $\mathcal{C}_{\text{app}}$ modifies the constrained stationarity to $\alpha\rho + \beta g -$
 515 $\mu b^{-2\phi} + \nu = 0$, whose twice-differentiated form is

$$\rho''(\phi) - \tau^2 \rho(\phi) = \gamma b^{-2\phi}, \quad (12)$$

516 with γ set by the Fisher multiplier's sign convention. The non-resonant general solution ($\tau \neq 2 \log b$)
 517 is

$$\rho_{\text{full}}(\phi) = C_1 \cosh(\tau\phi) + C_2 \sinh(\tau\phi) + \frac{\gamma}{(2 \log b)^2 - \tau^2} b^{-2\phi}. \quad (13)$$

518 At resonance, $\tau = 2 \log b$, the forcing profile $b^{-2\phi}$ coincides with one homogeneous mode. The
 519 particular solution is then

$$\delta\rho_F^{\text{part}}(\phi) = -\frac{\gamma}{2\tau} \phi e^{-\tau\phi}, \quad (14)$$

520 with the homogeneous constants again fixed by the boundary and mass conditions. All bounds
 521 below containing $(s^2 - \tau^2)^{-1}$ are non-resonant bounds and should be read with $|s^2 - \tau^2|^{-1}$; they
 522 are not uniform as $\tau \rightarrow 2 \log b$. EVQ-Cosh is the $\gamma=0$ homogeneous branch. The forced term is a
 523 controlled but nonzero residual; its magnitude and the corresponding warp-amplification caveat are
 524 bounded in §A.16.

525 A.3 Closed-form solution and geometric limit

526 The non-resonant ODE has the closed-form solution

$$\rho^*(\phi) = C_1 \cosh(\tau\phi) + C_2 \sinh(\tau\phi) + P b^{-2\phi}. \quad (15)$$

527 The first two terms span the homogeneous solution of $\rho'' - \tau^2 \rho = 0$, and the final term is a particular
 528 solution induced by the Fisher-weighted source $b^{-2\phi}$. This decomposition is the source of the main
 529 paper's "tether + pulse" interpretation.

530 To translate the density into RoPE frequencies, write the cumulative map

$$F(\phi) = \int_0^\phi \rho^*(s) ds, \quad (16)$$

531 normalize it to $[0, 1]$, and invert at the discrete quantiles $u_k = k/K$. In the pure-tether branch this
 532 inversion has the closed form

$$\phi_k(\tau) = 1 - \frac{1}{\tau} \operatorname{arcsinh}((1 - u_k) \sinh \tau). \quad (17)$$

533 To recover geometric RoPE, expand $\sinh \tau \approx \tau$ as $\tau \rightarrow 0$. Then

$$\phi_k(\tau) \rightarrow 1 - \frac{1}{\tau} \operatorname{arcsinh}((1 - u_k)\tau) \rightarrow u_k, \quad (18)$$

534 which is exactly the uniform quantile map.

535 This also shows why the high-frequency end remains effectively anchored: for $u_0 = 0$, one has
 536 $\phi_0(\tau) = 0$ exactly, so the leading inverse frequency remains $b^0 = 1$, matching geometric RoPE.

537 A.4 Why the pure-tether branch

538 The full stationary density contains three terms: a hyperbolic tether ($\cosh, \sinh$) and a particular so-
 539 lution $P b^{-2\phi}$. The practical EVQ method uses only the pure-tether branch, justified on two grounds.

540 **(1) Closed-form invertibility.** The pure-tether density $\rho(\phi) \propto \cosh(\tau(1 - \phi))$ admits an analyti-
 541 cally invertible CDF, yielding the closed-form warp $\phi_k(\tau)$. Including the $b^{-2\phi}$ term destroys this
 542 invertibility and requires numerical root-finding.

543 **(2) The forcing branch is globally small in L^1 /CDF transport, not pointwise.** Once $\gamma(b) =$
 544 $(\lambda_F/\alpha) \bar{\eta}_F (2 \log b)^2$ is derived (Appendix A.16, Eq. (45)), the $(s^2 - \tau^2)^{-1}$ denominator in the
 545 particular solution $\delta\rho_F^{\text{part}} = \gamma(b) q_s/(s^2 - \tau^2)$ is cancelled by the s^2 in $\gamma(b)$, so the forc-
 546 ing amplitude is *not* pointwise $O(1/\log^2 b)$. The true suppression is boundary-layer / L^1 -level:

547 $\|q_s\|_1 = (1 - e^{-s})/s = O(1/\log b)$, giving $\|\rho_{\text{full}} - \rho_{\text{EVQ}}\|_1 = O(1/\log b)$ and CDF-level error
 548 of the same order (Eq. (47)). The full solution improves over pure tether by 2–6% *on the offline*
 549 *collision/utility surrogate objective at unit-convention* $\lambda=1$ (vs. 24–92% improvement over geomet-
 550 ric on the same objective)—this is a surrogate gap, not a trained-PPL claim, and no main empirical
 551 claim relies on the forced solution. Furthermore, since inverse-CDF inversion can amplify CDF
 552 perturbations by $1/\inf \rho_\tau \sim \sinh \tau/\tau$, at large τ (e.g. $\tau=4$) the forced branch is best treated as a nu-
 553 merical extension rather than dismissed by the L^1 bound; the deployed pure-tether warp captures the
 554 dominant Geo→EVQ direction (24–92%) while remaining closed-form. The 99-run training sweep
 555 (Figure 4) is consistent with the pure-tether family matching the scaling-law prediction, subsuming
 556 geometric RoPE as its $\tau \rightarrow 0$ limit.

557 A.5 Why constant α (and not the continuum stationary-phase coefficient)

558 A natural objection asks why $\mathcal{C}_{\text{app}}$ uses a constant diagonal $\alpha\delta(\phi-\psi)$ rather than the ϕ -
 559 dependent coefficient implied by stationary-phase asymptotics of the exact kernel. With
 560 $\lambda_b := \ln b$ (distinct from the transport multiplier λ of §A.11), $D_0 := D(0)$, near $\psi=\phi$ a stationary-
 561 phase expansion of $K(\phi_1, \phi_2) = \int D(\Delta) \cos(\omega_1 \Delta) \cos(\omega_2 \Delta) d\Delta$ gives leading diagonal coefficient
 562 $\alpha_{\text{sp}}(\phi) = \pi D_0 / (L \lambda_b b^{-\phi}) = (\pi D_0 / L \lambda_b) b^\phi$, which grows exponentially in ϕ . The difference is not a
 563 higher-order correction:

$$\int_0^1 \rho^2(\phi) [\alpha_K - \alpha_{\text{sp}}(\phi)] d\phi = \frac{\|\rho\|_2^2}{2K} - \frac{\pi D_0}{L \lambda_b} \int_0^1 \rho^2(\phi) b^\phi d\phi, \quad (19)$$

564 which vanishes in the joint limit $K \rightarrow \infty$ and $L \lambda_b / b \rightarrow \infty$, but is $O(K^{-1} + (L \lambda_b)^{-1} \cdot b)$ otherwise. Re-
 565 placing α by $\alpha_{\text{sp}}(\phi)$ converts the homogeneous Euler–Lagrange ODE $\rho'' - \tau^2 \rho = 0$ into the variable-
 566 coefficient equation $-(\alpha_{\text{sp}}(\phi) \rho)'' + \beta \rho = 0$, whose closed-form solution is a modified-Bessel com-
 567 bination $\rho(\phi) \propto e^{-\lambda_b \phi} [A I_0(x(\phi)) + B K_0(x(\phi))]$ with $x(\phi) = (2r/\lambda_b) e^{-\lambda_b \phi/2}$, $r = \sqrt{\beta L \lambda_b / (\pi D_0)}$.
 568 This Bessel density has heavier right tail than cosh and, in some parameter regimes ($x_0 \rightarrow \infty$, $x_1 \rightarrow 0$),
 569 violates positivity at $\phi=1$ unless an active inequality constraint is added. We deliberately use constant
 570 α for three reasons. (i) *Closed-form CDF*: the cosh density admits the elementary inverse-CDF
 571 $\phi_k(\tau) = 1 - \tau^{-1} \text{arsinh}((1-u_k) \sinh \tau)$, which is the prerequisite for inverse-CDF channel quanti-
 572 zation; the Bessel density does not. (ii) *Manifest positivity*: $\rho_\tau \geq \tau / \sinh \tau > 0$ on $[0, 1]$ for all $\tau > 0$,
 573 while the variable- α Bessel solution can violate $\rho > 0$ at the right endpoint. (iii) *Discrete-grid fit, not*
 574 *continuum asymptotic*: paper-deployed $\alpha \approx 1/d_{\text{rot}}$ is fitted to K on the discrete $K \in \{16, 32\}$ grid
 575 (§A.11), and the resulting cosh allocation is functionally validated against the exact kernel by the
 576 24–92% collision reduction across 12 configurations (§A.6). $\mathcal{C}_{\text{app}}$ is therefore a deliberately discrete
 577 surrogate, not a claim about continuum-limit asymptotics; (19) is the precise statement of how the
 578 two diverge.

579 A.6 Surrogate quality: functional validation

580 The broadband surrogate $K_{\text{app}} = \alpha\delta + \beta \min$ is the theory’s sole approximation.
 581 We validate it not by pointwise comparison with the exact kernel $K(\phi_1, \phi_2) =$
 582 $\int D(\Delta) \cos(\omega(\phi_1) \Delta) \cos(\omega(\phi_2) \Delta) d\Delta$ — which is oscillatory (sinc-like) and cannot be well-
 583 approximated pointwise by a smooth two-parameter model — but by the *functional* test: does the
 584 EVQ allocation derived from K_{app} also reduce the collision objective under the true kernel K ?
 585 The 24–92% reduction reported below also serves as the operational forced-branch residual diagnos-
 586 tic: if the homogeneous (pure-tether) branch missed the dominant collision-reduction direction, the
 587 exact-kernel test would show no consistent improvement at the deployed τ .

588 Table 5 answers affirmatively. For a uniform distance prior $D(\Delta) = 1/L$ on $[0, L]$, we compute
 589 the exact kernel K at the discrete allocation points for both geometric and EVQ, then evaluate the
 590 normalized collision score $C = \sum_{i < j} K_{ij}^2 / (K_{ii} K_{jj})$ and effective spectral rank (exponential of
 591 the Shannon entropy of the eigenvalue distribution). We sweep across six training lengths ($L \in$
 592 $\{128, \dots, 4096\}$), three base values ($b \in \{10\text{K}, 100\text{K}, 500\text{K}\}$), and two channel counts ($K \in$
 593 $\{16, 32\}$), covering both text and video regimes.

594 EVQ reduces collision under the exact kernel in all 12 configurations (–24% to –92%), with effec-
 595 tive rank improvements of +24% to +570%. Two patterns emerge. First, the collision reduction
 596 monotonically increases as the geometric allocation becomes more degenerate: shorter L (more

Table 5: Functional surrogate validation across 12 configurations. EVQ (derived from K_{app}) reduces collision and increases effective rank under the *exact* kernel K in every tested regime. Collision reduction monotonically increases with extrapolation stress (shorter L , larger b).

Configuration	Collision C		Eff. rank	
	GEO	EVQ (Δ_C)	GEO	EVQ
<i>Text: $K=32$, $b=500K$, sweep L</i>				
$L=128$	226.3	17.7 (−92%)	2.5	17.0
$L=256$	192.6	27.6 (−86%)	3.0	13.9
$L=512$	163.2	43.3 (−73%)	3.6	10.8
$L=1024$	133.9	59.0 (−56%)	4.4	8.8
$L=2048$	109.0	67.5 (−38%)	5.4	8.0
$L=4096$	87.0	66.3 (−24%)	6.6	8.1
<i>Text: $K=32$, $L=2048$, sweep b</i>				
$b=10K$	34.8	19.2 (−45%)	13.0	17.3
$b=100K$	79.4	47.0 (−41%)	7.1	10.4
<i>Video: $K=16$, $L=32$, sweep b</i>				
$b=100$	22.6	14.2 (−37%)	4.8	6.4
$b=1K$	42.3	28.0 (−34%)	2.8	3.9
$b=10K$	55.4	39.8 (−28%)	2.2	2.9
$b=50K$	63.3	47.2 (−26%)	2.0	2.5

extrapolation stress at fixed τ^*) and larger b (more dead channels) both amplify the advantage, consistent with the dead-channel mechanism. Second, even in the most favorable regime for geometric RoPE ($L=4096$, where baseline effective rank is already 6.6), EVQ still improves collision by −24%, supporting the interpretation that the surrogate identifies a genuine structural improvement within the tested parameter range, not only in extreme regimes.

A.7 Exact higher-order displacement floor

The one-channel mid-displacement threshold admits an exact Taylor series in $1/K$ that is fully closed-form (no empirical calibration).

Proposition 2 (Higher-order floor). *For the EVQ-Cosh mapping with $K = d_{\text{rot}}/2$ channels, the N -channel mid-displacement threshold satisfies*

$$\tau_{\text{floor}}(N, K) = 4\sqrt{\frac{N}{K}} \left[1 + \frac{N}{2K} + \frac{241}{120} \left(\frac{N}{K} \right)^2 + O(K^{-3}) \right]. \quad (20)$$

All coefficients are derived from the Taylor series $\Delta\varphi(1/2, \tau) = -\tau^2/16 + \tau^4/256 + 17\tau^6/30720 + O(\tau^8)$, whose coefficients (−1/16, +1/256, +17/30720) are exact rationals from the expansion of $\text{arcsinh}(\sinh \tau/2)/\tau$ (Appendix script `theory_B_floor_higher_order.py`; verified symbolically).

Accuracy. Against the exact floor (obtained by numerically solving $|\Delta\varphi(1/2, \tau)| = N/K$): the leading term $4\sqrt{N/K}$ has relative error 3.85% at $K=16$ and 0.20% at $K=512$; the two-term expansion reduces this to 0.088% at $K=16$ and below 10^{-5} for $K \geq 128$. The formula is thus a closed-form refinement of the leading one-channel mid-displacement threshold, with all constants derived from the cosh geometry alone.

A.8 Waterbed envelope and corollary

Setup. With $\rho > 0$, $\int_0^1 \rho = 1$, $U(d\phi) = d\phi$, $P_\rho(d\phi) = \rho d\phi$, $\delta\rho = \rho - 1$, and $M_\rho(s) = \int_s^1 \delta\rho du$, define the waterbed functional

$$\mathcal{W}(\rho) := \alpha \chi^2(U\|P_\rho) + \beta \int_0^1 M_\rho(s)^2 ds, \quad \chi^2(U\|P_\rho) = \int_0^1 \frac{(1-\rho)^2}{\rho} d\phi. \quad (21)$$

619 The first term is the Pearson divergence of U from P_ρ (equivalently, $\int 1/\rho - 1$); the second is the
 620 integrated squared cumulative error, which arises from the min-kernel off-diagonal covariance in
 621 K_{app} .

622 **Proposition 3** (Waterbed inequality). *Let $D_B(\rho) = -\int_0^1 \log \rho(\phi) d\phi$ be the Burg entropy deficit.*
 623 *Then*

$$\mathcal{W}(\rho) \geq \alpha F_*(D_B(\rho)), \quad F_*(d) = \inf_{\rho > 0, \int \rho = 1, -\int \log \rho = d} \left[\int_0^1 \frac{1}{\rho} - 1 \right], \quad (22)$$

624 *with equality $\mathcal{W}(\rho) = 0$ if and only if $\rho \equiv 1$ (geometric RoPE). A readable corollary via Rényi*
 625 *monotonicity is*

$$\mathcal{W}(\rho) \geq \alpha(e^{D_B(\rho)} - 1). \quad (23)$$

626 *Proof sketch.* Since $\beta \int M_\rho^2 \geq 0$, $\mathcal{W}(\rho) \geq \alpha \chi^2(U \| P_\rho)$. For fixed $D_B(\rho) = d$, the sharp lower enve-
 627 lope of χ^2 over all densities is the Csiszár–Vajda f -divergence comparison F_* , whose extremizer is
 628 a two-level density $X = 1/\rho \in \{a, c\}$ with mass-normalization and entropy-deficit constraints. For
 629 the corollary, $1 + \chi^2(U \| P_\rho) = \int 1/\rho \mathbb{E}_U[\cdot] \geq e^{D_1(U \| P_\rho)} = e^{D_B(\rho)}$ by Rényi-2 $\geq$ Rényi-1. Equality
 630 at $\rho \equiv 1$ is immediate from both sides vanishing. $\square$

631 **EVQ evaluation.** For $\rho_\tau(\phi) = \tau \cosh(\tau(1-\phi)) / \sinh \tau$ (Eq. (25) below uses the un-normalized
 632 integral; cf. §A.17 “Normalization convention”),

$$D_B(\rho_\tau) = \log \frac{\sinh \tau}{\tau} - \frac{1}{\tau} \int_0^\tau \log \cosh y dy = \frac{\tau^4}{90} - \frac{8\tau^6}{2835} + O(\tau^8), \quad (24)$$

$$S_{\chi^2}(\rho_\tau) = \frac{\sinh \tau}{\tau^2} \arctan(\sinh \tau) - 1 = \frac{\tau^4}{45} - \frac{2\tau^6}{315} + O(\tau^8), \quad (25)$$

$$\int_0^1 M_{\rho_\tau}(s)^2 ds = \frac{2\tau^4}{945} + O(\tau^6). \quad (26)$$

633 Combining,

$$\mathcal{W}(\rho_\tau) = \left(\frac{\alpha}{45} + \frac{2\beta}{945} \right) \tau^4 + O(\tau^6), \quad \mathcal{W}(\rho_{\tau_*}) = O\left(\frac{d_{\text{head}}^4}{(L_{\text{eff}}^J)^2} \right). \quad (27)$$

634 The in-range surrogate cost for EVQ is $O(\tau^4)$; since $\tau_*^2 \asymp d_{\text{head}}^2 / L_{\text{eff}}^J$, the cost scales as
 635 $d_{\text{head}}^4 / (L_{\text{eff}}^J)^2$. This is the explicit form of “bounded in-range cost” cited in the main text.

636 A.9 Surrogate self-consistency theorem

637 The surrogate functional $\mathcal{J}[\rho]$ admits three exact identities that connect its two integral
 638 terms $T_1(\tau) = \int_0^1 \rho^2 d\phi$ and $T_2(\tau) = \int_0^1 \int_0^1 \rho(\phi) \rho(\psi) \min(\phi, \psi) d\phi d\psi$, where $\rho(\phi) =$
 639 $\tau \cosh(\tau(1-\phi)) / \sinh \tau$ is the normalized cosh density.

640 **Theorem 3** (Surrogate self-consistency). *For all $\tau > 0$:*

$$\tau^2 T_2(\tau) + T_1(\tau) = \tau \coth \tau. \quad (28)$$

641 *Proof.* The Green’s function $g(\phi) = \int_0^1 \rho(\psi) \min(\phi, \psi) d\psi$ satisfies $g''(\phi) = -\rho(\phi)$ with $g(0) = 0$,
 642 $g'(1) = 0$ (from the min kernel structure; see §A). Define $h(\phi) = \rho(0) - \rho(\phi)$. Then $h'' = -\rho''$,
 643 and under the pure-tether ODE $\rho'' = \tau^2 \rho$ this gives $h'' = -\tau^2 \rho$, with $h(0) = 0$ and $h'(0) = -\rho'(0)$.

644 Define $f(\phi) = \tau^2 g(\phi) - h(\phi)$. Then $f'' = \tau^2 g'' - h'' = -\tau^2 \rho - (-\tau^2 \rho) = 0$, so f is linear in ϕ .
 645 The boundary values give $f(0) = \tau^2 g(0) - h(0) = 0$ and $f'(0) = \tau^2 g'(0) + \rho'(0) = \tau^2 - \tau^2 = 0$
 646 (using $g'(0) = \int_0^1 \rho = 1$ and $\rho'(0) = -\tau^2$ from the pure-tether density). Hence $f \equiv 0$, yielding
 647 $\tau^2 g(\phi) = h(\phi) = \rho(0) - \rho(\phi)$, i.e.,

$$g(\phi) = \frac{\rho(0) - \rho(\phi)}{\tau^2}. \quad (29)$$

648 Integrating against ρ :

$$T_2 = \int_0^1 \rho(\phi) g(\phi) d\phi = \frac{1}{\tau^2} [\rho(0) \underbrace{\int_0^1 \rho}_{=1} - T_1] = \frac{\rho(0) - T_1}{\tau^2}. \quad (30)$$

649 Since $\rho(0) = \tau \cosh \tau / \sinh \tau = \tau \coth \tau$, rearranging gives (28). $\square$

650 An immediate corollary follows.

Corollary 1 (Closed-form T_2).

$$T_2(\tau) = \frac{\sinh 2\tau - 2\tau}{4\tau \sinh^2 \tau}. \quad (31)$$

651 This follows from substituting the known closed form $T_1(\tau) = \tau(1 + \sinh 2\tau/2\tau)/(2 \sinh^2 \tau)$ and
652 $\rho(0) = \tau \coth \tau$ into the theorem. Both identities are verified numerically to machine precision
653 ($< 10^{-15}$ relative error across $\tau \in [0.01, 20]$; code: `scripts/lambda_identity.py`).

654 The self-consistency theorem reveals that the surrogate’s two energy terms are not independent: they
655 are locked by the cosh density’s self-reproducing property under the min-kernel Green’s function.
656 This structural rigidity is why the pure-tether solution (§A) performs so well despite being a single-
657 parameter family.

658 A.10 Cross-validation of λ

659 The operating-point rule $\tau^* = c \cdot d_{\text{head}}/\sqrt{L}$ is analyzed at two levels.

660 **Collision-score level.** We minimize the exact-kernel collision score $C(\tau)$ for each of 9 configura-
661 tions ($d_{\text{head}} \in \{32, 64, 128\}$, $L \in \{256, 512, 1024\}$) and compute $c_{\text{coll}} = \tau_{\text{coll}}^*/(d_{\text{head}}/\sqrt{L})$. The
662 resulting $c_{\text{coll}} \approx 1.17$ is nearly constant (CV: 0.11%); leave-one-out cross-validation gives max pre-
663 diction error $< 0.4\%$ (Table 6). This near-zero spread reflects the smoothness of the collision score
664 as a function of τ , not PPL variability.

Table 6: Collision-score validation of the operating-point prefactor c in $\tau^* = c \cdot d_{\text{head}}/\sqrt{L}$. Column $c_{\text{pred}}(L, b) = \sqrt{45} Q_1(L, b)$ is the theory prediction under the $\lambda=1$ unit convention (Appendix A.18); column $c_{\text{coll}} = \tau_{\text{coll}}^*/(d_{\text{head}}/\sqrt{L})$ is the exact-kernel collision-score minimizer. The ratio $c_{\text{coll}}/c_{\text{pred}} = 0.981 \pm 0.003$ (CV 0.28%, 9 configurations) is near-unity, supporting surrogate-exact curvature agreement at the 2% level; implied curvature ratio $\lambda_\infty = 0.96 \pm 0.005$. Leave-one-out cross-validation on c_{coll} gives max prediction error 0.008 ($< 0.4\%$ relative, RMSE 0.004), so the near-zero CV reflects the smoothness of the collision score, not PPL variability (PPL sweep best points exhibit CV $\approx 14\%$). The deployed formula uses $\lambda=1$, which falls inside the $< 1\%$ PPL basin across all 27 configurations.

d_{head}	L	Q_1	c_{pred}	c_{coll}	Ratio	Error
32	256	0.03145	1.190	1.170	0.984	+0.002
32	512	0.03192	1.199	1.174	0.979	−0.005
32	1024	0.03159	1.192	1.170	0.981	+0.001
64	256	0.03145	1.190	1.170	0.984	+0.004
64	512	0.03192	1.199	1.170	0.976	+0.002
64	1024	0.03159	1.192	1.170	0.981	+0.002
128	256	0.03145	1.190	1.171	0.985	−0.004
128	512	0.03192	1.199	1.172	0.978	−0.008
128	1024	0.03159	1.192	1.170	0.981	+0.004
Mean $c_{\text{coll}}/c_{\text{pred}} = 0.981 \pm 0.003$ (CV 0.28%); implied $\lambda_\infty = 0.96 \pm 0.005$						

665 **PPL-sweep level.** The corresponding training sweeps (9 configurations $\times$ 3+ seeds) identify the
666 best observed PPL sweep point for each configuration. The resulting $\lambda_{\text{PPL}} = \tau_{\text{PPL}}^*/(d_{\text{head}}/\sqrt{L})$
667 has mean ≈ 1.19 with broader scatter (CV $\approx 14\%$), because PPL depends on training noise, finite

data, and the discrete sweep grid. The deployed formula uses $\lambda=1$ (i.e. $\tau = d_{\text{head}}/\sqrt{L}$ without a multiplicative constant), which consistently falls inside the flat PPL basin: the PPL gap between the formula and the best observed grid point is $<1\%$ across all 27 tested configurations.

Basin flatness. The collision-score landscape $C(\tau)$ for a representative configuration ($d_{\text{head}}=64$, $L=512$, $b=500\text{K}$) is flat near the optimum: $\pm 20\%$ deviation from τ_{coll}^* incurs $<1\%$ collision increase. This flatness is why the deployed bare formula $\tau = d_{\text{head}}/\sqrt{L}$ (i.e. omitting the $\sqrt{45 Q_1(L, b)}$ prefactor for practical simplicity, so $c_{\text{deployed}}=1$) sits $\approx 15\%$ below $c_{\text{coll}}=1.171$ —inside the $\pm 20\%$ collision basin—and consequently inside the $<1\%$ PPL basin observed in trained sweeps. The full theoretical prediction $c_{\text{pred}}=\sqrt{45 Q_1}\approx 1.19$ matches c_{coll} to 2% (see “Numerical validation” below); the practical and theoretical prefactors differ, but both land inside the same flat basin.

b -dependence of c_{pred} . The basin width above is verified at the representative $(d_{\text{head}}, L, b)=(64, 512, 500\text{K})$. The theoretical prefactor $c_{\text{pred}}(L, b)=\sqrt{45 Q_1(L, b)}$ varies modestly with b : at $b=500\text{K}$ across $L\in[128, 4096]$, $c_{\text{pred}}\in[1.14, 1.20]$ and the deployed $c=1$ deviates 12–17% (inside the $\pm 20\%$ collision basin); at $b=100\text{K}$ the range is $[1.06, 1.20]$ and deployed $c=1$ stays inside the basin throughout; at $b=10\text{K}$, c_{pred} drops to 0.94 at $L=2048$ (+7% deviation, still inside the basin) and 0.80 at $L=4096$ (+25% deviation, marginally *outside* the $\pm 20\%$ basin). All primary experiments use $b\geq 500\text{K}$ where the bare formula stays inside the basin; at small b and large L , reverting to the explicit $c_{\text{pred}}(L, b)=\sqrt{45 Q_1(L, b)}$ prefactor is appropriate.

Why λ is a normalization convention, not a free parameter. The Lagrange multiplier λ in the softmax-transport functional $\mathcal{F}(\tau) = \frac{1}{2} S_{\chi^2}(\tau) - \lambda U(\tau, L)$ has a closed-form interpretation. We use d_{head} (not the number of RoPE frequency pairs $K = d_{\text{head}}/2$) as the stiffness normalization throughout both main text and appendix; all factors of two arising from the RoPE pair structure are absorbed into the $\mathcal{O}(1)$ convention-dependent prefactor. We adopt the Dirichlet-energy convention in which S_{χ^2} enters $\mathcal{F}$ with a $\frac{1}{2}$ prefactor, so that under the small- τ expansion $\frac{1}{2} S_{\chi^2}(\tau) = \tau^4/(90 d_{\text{head}}) + \mathcal{O}(\tau^6)$ (using $S_{\chi^2}(\tau) = \tau^4/(45 d_{\text{head}})$ from §A.12); this matches the numeric constant 45 appearing in Proposition 1. Similarly, $U(\tau, L) = (d_{\text{head}}/L)[Q_0 + \tau^2 Q_1(L, b)] + \mathcal{O}(\tau^4)$, where $Q_1(L, b) = \int_0^1 \eta(\phi) q(Lb^{-\phi}) d\phi$ is an explicit integral of known functions. All numerical factors below are reported under this $\frac{1}{2}S$ convention; a reader using an un-normalized S would absorb a factor of 2 into λ . The balance equation $\partial\mathcal{F}/\partial\tau = 0$ yields

$$\tau_*^2 = 45 \lambda Q_1(L, b) \cdot \frac{d_{\text{head}}^2}{L}.$$

Splitting Q_1 into an active-channel contribution ($q \approx 1/2$ for $Lb^{-\phi} \gg 1$, i.e. $\phi \in [0, \phi^*]$ with $\phi^* = \ln L / \ln b$) and a dead-channel contribution ($q(x) = x^4/45 + \mathcal{O}(x^6)$ for $x = Lb^{-\phi} \ll 1$ on $\phi \in [\phi^*, 1]$; the leading Taylor coefficient is $x^4/45$, not $x^2/45$) gives the closed form $Q_1(L, b) \approx \frac{1}{12} [1 - (1 - \phi^*)^3 - \phi^*]$ to leading order, with the dead-channel correction exponentially small. To avoid symbol reuse, we reserve λ for the utility-weight normalization in $\mathcal{F}(\tau) = \frac{1}{2} S_{\chi^2}(\tau) - \lambda U(\tau, L)$ and write the multiplicative operating-point prefactor as c , so $\tau^* = c \cdot d_{\text{head}}/\sqrt{L}$. Combining the expansions above gives the theoretical prefactor $c_{\text{pred}}(L, b; \lambda) = \sqrt{45 \lambda Q_1(L, b)}$. Under the deployed convention $\lambda=1$, this is a posterior-consistency prediction (computable from (L, b)), not a per-experiment fitted hyperparameter; the residual $\mathcal{O}(1)$ offset is absorbed into the flat PPL basin rather than treated as a tunable parameter.

Probabilistic interpretation of $q(x)$. The function $q(x)=1/2+\sin(2x)/(4x)-(\sin x/x)^2$ is not a postulated form; it is the cosine phase-variance under a uniform separation prior $t \sim \text{Unif}[0, 1]$:

$$q(x) = \text{Var}_{t \sim U[0,1]}[\cos(xt)] = \mathbb{E}_t[\cos^2(xt)] - (\mathbb{E}_t[\cos(xt)])^2,$$

since $\mathbb{E}_t[\cos^2(xt)] = 1/2 + \sin(2x)/(4x)$ and $\mathbb{E}_t[\cos(xt)] = \sin(x)/x$. Operationally, $q(Lb^{-\phi})$ measures how much information channel ϕ retains about a uniformly distributed relative position over $[0, L]$: in the dead-channel tail ($Lb^{-\phi} \ll 1$) the cosine is nearly constant and $q \rightarrow 0$; in the active region ($Lb^{-\phi} \gg 1$) the phase is well-sampled and $q \rightarrow 1/2$. Thus $Q_1(L, b)$ weights the cosh-driven density perturbation $\eta(\phi)$ by the per-channel phase-variance budget, and $q(x)$ is a self-contained signal-processing quantity rather than an unmotivated kernel.

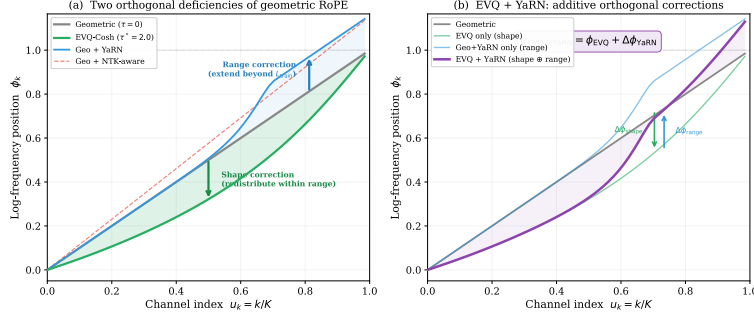


Figure 3: **EVQ and YaRN correct orthogonal deficiencies of geometric RoPE.** EVQ bends the allocation *below* the diagonal (shape); YaRN shifts it *above* (range). Their combination is additive. Setup: base = 10,000, $d_{\text{head}}=128$, $L_{\text{train}}=4096$, $4\times$ extrapolation.

A.11 Scaling analysis of τ^* : from surrogate to operating point

The operating-point rule

$$\tau^* = \frac{d_{\text{head}}}{\sqrt{L}} \quad (32)$$

is supported across 99 trained models spanning 27 validation settings (Figure 4); systems and MLA claims rely on 3-seed anchors within this grid, while the PE-dominant diagnostic and 1–2-seed rows extend scope. Here we analyze to what extent the broadband surrogate predicts this formula, and identify the gap that remains.

Numerical characterization of surrogate coefficients. We fit $K_{\text{app}}(\phi_i, \phi_j) = \alpha \delta_{ij}/\Delta\phi + \beta \min(\phi_i, \phi_j)$ to the exact collision kernel $K(\phi_i, \phi_j) = \frac{1}{2L} \left[\frac{\sin((\omega_1 - \omega_2)L)}{\omega_1 - \omega_2} + \frac{\sin((\omega_1 + \omega_2)L)}{\omega_1 + \omega_2} \right]$ evaluated at $K = d_{\text{head}}/2$ uniformly spaced channel positions in $[0, 1]$. Sweeping across $L \in \{128, \dots, 4096\}$, $d_{\text{head}} \in \{32, 64, 128, 256\}$, and $b \in \{10K, 100K, 500K\}$ yields:

$$\alpha \approx \frac{1}{2K} = \frac{1}{d_{\text{head}}}, \quad \beta \sim L^{-0.22}, O(1). \quad (33)$$

The diagonal coefficient α is set by the kernel’s diagonal value ($K(\phi, \phi) \approx 1/2$ for most channels) times the channel spacing $\Delta\phi = 1/K$. Crucially, α depends on d_{head} through the discretization ($K = d_{\text{head}}/2$) and is nearly independent of L . The off-diagonal coefficient β has weak power-law decay with L and negligible d_{head} dependence.

What the surrogate predicts. Substituting into $\tau_{\text{surr}} = \sqrt{\beta/\alpha}$:

$$\tau_{\text{surr}} = \sqrt{\beta \cdot d_{\text{head}}} \sim \sqrt{d_{\text{head}}} \cdot L^{-0.11}. \quad (34)$$

This correctly captures the d_{head} dependence: $\alpha \propto 1/d_{\text{head}}$ implies $\tau \propto \sqrt{d_{\text{head}}}$, matching the direction and monotonicity of the observed scaling. However, the L -scaling is too weak: $L^{-0.11}$ vs. the observed $L^{-0.5}$.

What the surrogate does and does not supply. Under the probability-density convention used throughout the paper, $\int_0^1 \rho = 1$; the number of rotary channels enters through the deployed discrete grid and through the fitted diagonal coefficient $\alpha \approx 1/d_{\text{rot}}$ (for standard MHA, $d_{\text{rot}} = d_{\text{head}}$). Thus the broadband surrogate supplies the cosh allocation family and the finite-channel normalization direction, but it does not by itself derive the deployed operating rule $\tau = d_{\text{eff}}/\sqrt{L}$.

The remaining $L^{-1/2}$ structure comes from the separate post-softmax transport calculation in Proposition 1: the diffuse softmax Jacobian contributes the $1/L$ curvature factor, and balancing the resulting $O(\tau^2)$ utility against the Pearson- $\chi^2 O(\tau^4)$ stiffness gives the structural scaling

$$\tau_*^2 \propto \frac{d_{\text{eff}}^2}{L}. \quad (35)$$

742 The practical rule $\tau = d_{\text{eff}}/\sqrt{L}$ is therefore a semi-analytic operating-point selector with a
743 convention-dependent $\mathcal{O}(1)$ prefactor, validated inside the observed flat basin, not a global optimum
744 of either the full trained-transformer objective or the exact oscillatory kernel.

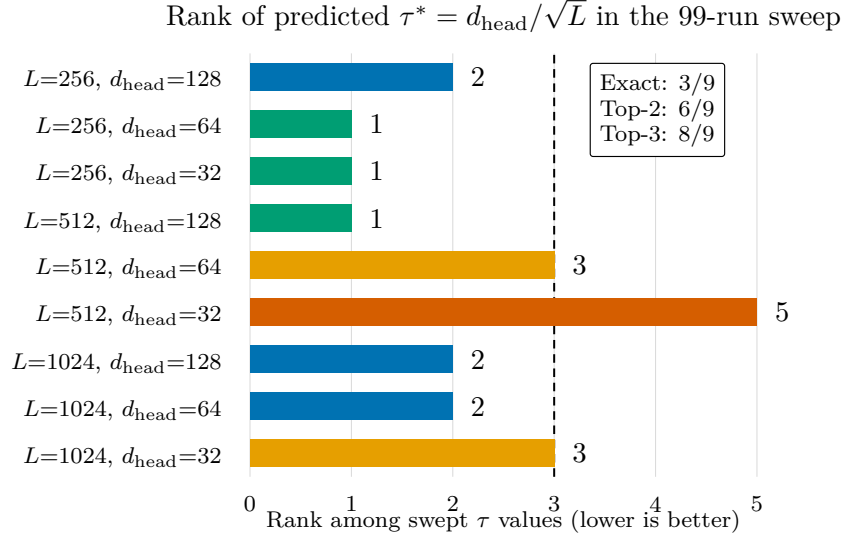


Figure 4: **Rank of the predicted operating point in the 99-run, 3-seed scaling-law grid.** Across the nine (L, d_{head}) configurations shown, the predicted $\tau^* = d_{\text{head}}/\sqrt{L}$ is exact-best in 3/9, top-2 in 6/9, and top-3 in 8/9. This supports a basin-selector interpretation rather than a sharp-optimum claim; Table 7 separately lists the broader 27-setting evidence grid and its seed tiers.

745 **Configuration breakdown.** Table 7 decomposes the 27 tested configurations by architecture, seed
746 count, and evidence tier.

Table 7: Breakdown of the 27 configurations in the tested grid for $\tau^* = d_{\text{head}}/\sqrt{L}$. Systems and MLA claims rely on 3-seed primary rows; the PE-dominant row and 1–2-seed rows extend scope with the stated seed count. The $0.53\times$ modality correction for video DiT is empirical (§B.6).

Architecture	Scale	d_{head}	L_{train}	Seeds	τ^* variant	Tier
MHA (text)	50M	64	2048	1	base	Robustness
MHA (text)	125M	64	128	1–3	base	Primary
MHA (text)	125M	64	2048	1	base	Robustness
MHA (text)	454M	64	128	1–3	base	Primary
MHA (text)	454M	64	256	3	base	Primary
MHA (text)	454M	64	512–2048	3	base	Primary
MHA (text)	454M	64	2048	3	base	Primary
MHA (text)	750M	64	4096	1	base	Supporting
MLA	432M	128	8192	3	base	Primary
DiT (video)	129M	16	32	2	$0.53\times$	Supporting
DiT (video)	382M	16	32	1	$0.53\times$	Supporting
LoRA	8B	128	8192	1	base	Supporting

747 A.12 Stiffness functional derivation

748 The softmax transport objective in Proposition 1 requires choosing a stiffness functional $S(\tau)$
749 measuring the cost of departing from uniform density. We test a one-parameter family $S_p(\tau) =$
750 $\frac{1}{d_{\text{head}}} \int_0^1 (\rho_\tau - 1)^2 / \rho_\tau^p d\phi$ using exact (non-Taylor) integrals of the normalized cosh density $\rho_\tau(\phi) =$
751 $\tau \cosh(\tau(1-\phi)) / \sinh \tau$.

Table 8: L -exponent γ (defined by $\tau_* \propto L^{-\gamma}$) under different stiffness functionals, computed via exact numerical optimization over $L \in [128, 4096]$. Target: $\gamma = 0.500$.

Stiffness	p	γ	Note
L^2 norm $\int (\rho-1)^2$	0	0.626	Original choice
Geometric mean	0.5	0.561	
Exponent-matched diagnostic	0.80	0.498	Sensitivity analysis, not used in deployment
Pearson χ^2	1.0	0.465	Channel-load variance penalty (axioms)
KL divergence	—	0.710	Worse than L^2
Reverse KL	—	0.581	
Jeffreys	—	0.621	$\approx L^2$
Rényi ($\alpha=2$)	—	0.807	

Why χ^2 . Two independent arguments favor $p=1$. (1) *Attention-load asymmetry*: each frequency channel at density $\rho(\phi)$ bears per-channel attention load $\propto 1/\rho$; weighting the squared departure by this load yields $\int (\rho-1)^2/\rho = \chi^2$. (2) *Discrete-grid effect*: the continuous L^2 integral, evaluated on EVQ’s non-uniform cosh-quantile grid, naturally acquires a $1/\rho$ Jacobian factor, producing discrete χ^2 .

Closed form. For $p=1$, the stiffness admits

$$S_{\chi^2}(\tau) = \frac{1}{d_{\text{head}}} \left[\frac{\sinh \tau \cdot \arctan(\sinh \tau)}{\tau^2} - 1 \right], \quad (36)$$

verified numerically to relative error $< 7 \times 10^{-6}$.

Exponent-matched diagnostic. As a sensitivity analysis, one can ask which member of the S_p family would match the target exponent $\tau_* \propto 1/\sqrt{L}$ exactly over this grid. The resulting $S_{\text{derived}}(\tau)$ lies near $p \approx 0.85$ (high-resolution sweep; consistent with 0.80 from the coarser grid), close to but distinct from the Pearson χ^2 choice $p=1$. We do not use this as a derivation of the stiffness; it is reported as sensitivity analysis. The full sweep is reproduced by scripts/analysis/verify_stiffness_and_regime.py (pure NumPy, ~ 60 s).

A.13 Training-regime dependence and the LoRA phase transition

Pretrained weights create a *coupling stiffness* that penalizes frequency redistribution. This subsection gives a phenomenological model calibrated by the rank sweep, not a first-principles derivation: the EVQ warp actually perturbs frequencies at order τ^2 (Eq. (38)), so the τ^2 coupling below is an *effective* single-scale surrogate for the leading non-perturbative interaction with frozen pretrained Q/K blocks, not the literal leading term of the small- τ expansion. Treating the frozen-weight mismatch as producing an aggregate logit variance $\text{Var}(\delta\ell) \propto \tau^2$, the total stiffness becomes

$$S_{\text{total}}(\tau; r) = S_{\chi^2}(\tau) + \Lambda_0 \left(1 - \frac{r}{K} \right) \frac{\tau^2}{d_{\text{head}}}, \quad (37)$$

where r is the LoRA rank, $K = d_{\text{head}}/2$, and $\Lambda_0 \gg 1$ for well-trained models. In this phenomenological model the coupling term dominates at small τ , and the system exhibits a *near-phase-transition* in effective viability: $\tau_* \rightarrow 0$ for small r/K and τ_* jumps to an $\mathcal{O}(1)$ value once r is large enough.

Numerical verification under this phenomenological model shows the transition is *apparent-sharp*: for LLaMA-8B parameters ($d_{\text{head}}=128$, $K=64$), sweeping LoRA rank over the variational landscape shows $\tau_*=0$ for all $r \leq 48$ ($r/K \leq 0.75$), then τ_* jumps to 1.41 at $r=64$ ($r/K=1.0$). The rank at which this occurs, $r \approx K$, is the value observed in this sweep; we do not claim an exact first-principles derivation of $r_c = K$, and the specific threshold depends on Λ_0 .

For LLaMA-8B with LoRA $r=16$: the adaptation ratio $r/K = 25\%$ leaves 48 of 64 frequency channels frozen. At $\tau=1.414$, approximately 8 channels are displaced, but the 48 frozen channels cannot adapt, creating systematic attention logit mismatch and driving PPL from 11.8 to 77.1 (this PPL value is reproduced after calibrating Λ_0 on the same observation, so it is a calibration check of

the phenomenological model rather than an out-of-sample prediction). The qualitative implication—LoRA with $r \gtrsim K$ should restore EVQ viability—is independent of this calibration and is consistent with the rank sweep in Table 24.

SFT coupling decay. Under full-parameter SFT, the coupling decays exponentially: $\tau_{\text{SFT}}^*(T) \approx \tau_{\text{pt}}^* / \sqrt{1 + \Lambda_0 e^{-T\eta\sigma}}$. Recovery to 70% of τ_{pt}^* requires $T\eta\sigma \approx 5$; full recovery ($> 95\%$) requires $T\eta\sigma \approx 10$.

A.14 Mechanism isolation: collision-only ablation

To verify that the collision-reduction mechanism identified by the surrogate theory is the primary driver of EVQ’s improvement, we perform a diagnostic experiment that isolates the collision objective from the full training loss.

We define the off-diagonal energy $E_{\text{off}} = \sum_{i < j} K_{ij}^2 / (K_{ii} K_{jj})$ under the exact RoPE kernel (not the surrogate), and optimize the EVQ allocation to minimize E_{off} directly (“collision-only”), without any language modeling. Table 9 compares geometric, EVQ at theory-predicted τ^* , and collision-optimized allocations across 36 configurations ($d_{\text{head}} \in \{32, 64, 128\}$, $L \in \{256, \dots, 8192\}$, $b \in \{10\text{K}, 500\text{K}\}$).

Table 9: Mechanism isolation: off-diagonal energy E_{off} and mutual coherence μ across representative configurations. Column “Oracle” reports a numerical search (collision-only objective, not a global minimizer; see caveat below). Geo and EVQ columns are evaluated at fixed allocations; −65% annotations are computed against the Geo baseline.

Configuration	$E_{\text{off}} (\downarrow)$			Mutual coherence μ	
	GEO	EVQ(τ^*)	Oracle	GEO	Oracle
$b=10\text{K}$					
$d=32, L=512$	653	516	220 (−66%)	.999	.708
$d=64, L=512$	2887	1533	1000 (−65%)	.999	.814
$d=128, L=512$	12120	3958	4283 (−65%)	.999	.973
$b=500\text{K}$					
$d=32, L=512$	850	690	286 (−66%)	1.00	.862
$d=64, L=512$	3715	2069	1305 (−65%)	1.00	.995
$d=128, L=2048$	16473	8899	5574 (−66%)	1.00	.998

Two consistent findings emerge across configurations, plus one caveat. First, both EVQ and the oracle search reduce E_{off} substantially relative to geometric in all tested configurations, and mutual coherence μ drops from ≥ 0.99 under geometric to 0.55–0.99 under the oracle—EVQ breaks the near-degeneracy of the geometric kernel in the same direction the oracle does. Second, the oracle capture fraction (defined as $(E_{\text{Geo}} - E_{\text{EVQ}}) / (E_{\text{Geo}} - E_{\text{Oracle}})$) varies across configurations, so we report the individual EVQ and oracle columns rather than a single capture statistic. *Caveat on the oracle column.* The oracle is a numerical collision-only search and is not guaranteed to find the global minimum; in the $d=128, L=512, b=10\text{K}$ row, EVQ at the theory-predicted τ^* achieves a lower E_{off} than the oracle search, indicating the search missed the true minimum at that configuration. We read the oracle numbers as a directional lower bound rather than as a tight global optimum.

A.15 Effective window L_{eff}^J via softmax-Jacobian Rayleigh quotient

From density perturbation to RoPE frequency shift. The EVQ pure-tether CDF is $F_\tau(\phi) = 1 - \sinh(\tau(1-\phi)) / \sinh \tau$, with inverse-CDF warp $\Phi_\tau(u) = 1 - \tau^{-1} \text{arsinh}((1-u) \sinh \tau)$. Setting $\theta = \tau^2$ and expanding $\rho_\tau(\phi) = 1 + \theta a(\phi) + O(\theta^2)$ with $a(\phi) = (1-\phi)^2/2 - 1/6$, the quantile displacement takes the closed form

$$\Phi_\tau(u) = u + \tau^2 h(u) + O(\tau^4), \quad h(u) = - \int_0^u a(v) dv = -\frac{1}{6} u(1-u)(2-u), \quad (38)$$

so RoPE frequencies perturb as $\omega_k(\tau) = \omega_k^{(0)} [1 - \tau^2 (\log b) h(u_k) + O(\tau^4)]$ with $\omega_k^{(0)} = b^{-u_k}$. Hence $\dot{\omega}_k := \partial_\theta \omega_k|_{\tau=0} = -(\log b) \omega_k^{(0)} h(u_k)$.

817 **Explicit RoPE logit transport direction.** Fix a layer/head and query i ; write each 2D RoPE
818 pair as $q_{i,k}^{\mathbb{C}} = q_{i,2k-1} + iq_{i,2k}$, $k_{j,k}^{\mathbb{C}} = k_{j,2k-1} + ik_{j,2k}$, and $\alpha_{ij,k} = q_{i,k}^{\mathbb{C}} \overline{k_{j,k}^{\mathbb{C}}} = C_{ij,k} +$
819 $iD_{ij,k}$. With relative position r_{ij} and attention scale s_{att} , the RoPE logit is $z_{ij}^{\text{rope}}(\tau) =$
820 $s_{\text{att}}^{-1} \text{Re} \sum_k \alpha_{ij,k} \exp(ir_{ij}\omega_k(\tau))$. The leading logit transport direction $g_i(j) := \partial_\theta z_{ij}^{\text{rope}}|_{\tau=0}$ has
821 the closed form

$$g_i(j) = \frac{(\log b) r_{ij}}{s_{\text{att}}} \sum_{k=1}^K \omega_k^{(0)} h(u_k) \left[C_{ij,k} \sin(r_{ij}\omega_k^{(0)}) + D_{ij,k} \cos(r_{ij}\omega_k^{(0)}) \right]. \quad (39)$$

822 This is the Fréchet derivative of the logit with respect to the EVQ allocation perturbation at fixed
823 activations. A finite- τ implementation that avoids first-order phase expansion at large $r_{ij}\omega$ is

$$g_i^{(\tau)}(j) = \frac{1}{s_{\text{att}}\tau^2} \text{Re} \sum_{k=1}^K \alpha_{ij,k} \left[\exp(ir_{ij}b^{-\Phi_\tau(u_k)}) - \exp(ir_{ij}b^{-u_k}) \right], \quad (40)$$

824 with $g_i(j) = \lim_{\tau \rightarrow 0} g_i^{(\tau)}(j)$.

825 **Primary definition: Rayleigh quotient of the softmax Jacobian.** For the trained attention distri-
826 bution $p_i(j)$ on keys $j \in \mathcal{K}_i$, $J_{\text{sm}}(p_i) = \text{diag}(p_i) - p_i p_i^\top$ satisfies $g_i^\top J_{\text{sm}}(p_i) g_i = \text{Var}_{j \sim p_i}[g_i(j)] =$
827 $\frac{1}{2} \sum_{j,k} p_i(j) p_i(k) (g_i(j) - g_i(k))^2$. Since $J_{\text{sm}}(p_i) \mathbf{1} = 0$, we center g_i by the key-wise mean operator
828 P_i . The effective length is then defined by the *ratio-of-means*

$$\frac{1}{L_{\text{eff}}^J} = \kappa_{\text{att}} = \frac{\mathbb{E}_{\ell,h,i} [g_i^\top J_{\text{sm}}(p_i) g_i]}{\mathbb{E}_{\ell,h,i} [\|P_i g_i\|_2^2]}, \quad (41)$$

829 which is the directional post-softmax curvature in the RoPE transport direction, averaged with the
830 natural utility weight $\|P_i g_i\|_2^2$. We prefer (41) over an average of per-query ratios because it avoids
831 noise amplification when $\|P_i g_i\|_2$ is small. Substituting $1/L$ by $1/L_{\text{eff}}^J$ in Proposition 1 gives

$$\begin{aligned} \tau_*^2 &= 45 \lambda Q_1^{\text{eff}}(L, b) \frac{d_{\text{head}}^2}{L_{\text{eff}}^J}, & \tau^* &\approx \frac{d_{\text{head}}}{\sqrt{L_{\text{eff}}^J}}, \\ \frac{d \log \tau^*}{d \log L_{\text{train}}} &= -\frac{1}{2} a(L_{\text{train}}), & a &:= \frac{d \log L_{\text{eff}}^J}{d \log L_{\text{train}}}. \end{aligned} \quad (42)$$

832 The paper’s $-1/2$ exponent is the case $a=1$ (diffuse growth); $a \rightarrow 0$ corresponds to saturation $L_{\text{eff}}^J \rightarrow$
833 $L_* < \infty$.

834 **Entropy proxy and its limits.** A cheap diagnostic is the harmonic mean of per-query entropy
835 lengths e^{H_i} , where $H_i = -\sum_j p_i(j) \log p_i(j)$:

$$\frac{1}{L_{\text{eff}}^H} = \mathbb{E}_{\ell,h,i} [e^{-H_i}]. \quad (43)$$

836 If each active window is approximately flat on support W_i of size m_i and g_i has energy con-
837 centrated on W_i , then $e^{-H_i} = 1/m_i$ and $g_i^\top J_{\text{sm}}(p_i) g_i / \|P_{W_i} g_i\|_2^2 = 1/m_i$, giving $L_{\text{eff}}^J = L_{\text{eff}}^H$.
838 The proxy and theoretical object differ in two regimes. First, if attention has nonuniform masses
839 $p_j = (1 + \varepsilon_j)/m$ and g_i covaries with ε_j , then the Rayleigh quotient picks up the first-order align-
840 ment term $\sum_j \varepsilon_j g_j^2 / \sum_j g_j^2$ which e^{-H} lacks. Second, for near-one-hot attention ($H_i \rightarrow 0$), $e^{-H_i} \rightarrow 1$
841 would suggest “effective length ≈ 1 ”, whereas $J_{\text{sm}}(p_i) \rightarrow 0$: the softmax saturates and post-softmax
842 sensitivity vanishes, so L_{eff}^J reads sharp attention as low-transport rather than short-but-sensitive. We
843 therefore treat L_{eff}^J as the theoretical quantity and L_{eff}^H as a cheap flat-window proxy.

844 **Perturbative self-consistency of Proposition 1.** Within the same perturbative softmax-transport
845 model, the replacement $1/L \mapsto 1/L_{\text{eff}}^J$ in Proposition 1 is a model-based extension of the diffuse-
846 baseline calculation rather than an empirical shortcut. Writing the log-partition function $A(z) =$
847 $\log \sum_j e^{z_j}$ and $p_\epsilon = \text{softmax}(z + \epsilon g)$, one has $D_{\text{KL}}(p \| p_\epsilon) = A(z + \epsilon g) - A(z) - \epsilon p^\top g =$
848 $\frac{1}{2} \epsilon^2 g^\top J_{\text{sm}}(p) g + R_3$ with $|R_3| \leq C_3 |\epsilon|^3 \|g\|_\infty^3$. Since the second derivative of A is $J_{\text{sm}}(p) =$

diag(p) - $pp^\top$, the directional curvature $g^\top J_{\text{sm}}(p)g = \text{Var}_{j \sim p}[g_j]$ depends only on the distribution p . For $p_0 = 1/L$ this reduces to $\|Pg\|^2/L$, the unique origin of the $1/L$ factor in the diffuse-baseline derivation. Substituting $1/L \mapsto \mathbb{E}[g^\top J_{\text{sm}}(p)g]/\mathbb{E}[\|Pg\|^2] = 1/L_{\text{eff}}^J$ and re-running the small- τ variational balance $F(\tau) = \frac{1}{2}S_{\chi^2}(\tau) - \lambda U(\tau, p)$ gives $\tau_*^2 = 45\lambda Q_1^J(p, b) d_{\text{head}}^2/L_{\text{eff}}^J + O(d_{\text{head}}^4/(L_{\text{eff}}^J)^2)$, with the residual R_3 absorbed into higher-order terms under the assumption $|\tau^2| \cdot \max_j |g_j|$ small. Thus, under the stated perturbative smallness condition, the same structural balance admits the trained-attention substitution; direct measurement of L_{eff}^J is treated as a falsifiable check rather than a theorem for arbitrary trained attention distributions.

Validation protocol. No retraining is required for the proxy test. On existing 16K or 32K checkpoints, log attention matrices on $\sim 10^3$ sequences and compute (43). For the theoretical object, evaluate (39) or (40) on the same batch and substitute into (41). A falsifying measurement would be either $L_{\text{eff}}^J \approx L_{\text{train}}$ with a saturating empirical optimum, or $L_{\text{eff}}^J \rightarrow L_* < \infty$ with $\tau^* \propto L_{\text{train}}^{-1/2}$ continuing to hold at $L_{\text{train}} \geq 16\text{K}$.

A.16 Pure-tether branch: Fisher-forcing coefficient and residual bound

Setup. The full stationary ODE under $\mathcal{C}_{\text{app}}$ plus a Fisher utility is $\rho'' - \tau^2 \rho = \gamma b^{-2\phi}$. Writing $s = 2 \log b$ and $q_s(\phi) = e^{-s\phi}$, the pure-tether branch $\rho_{\text{EVQ}}(\phi) = \tau \cosh(\tau(1-\phi))/\sinh \tau$ solves the homogeneous problem; the full solution is $\rho_{\text{full}} = \rho_{\text{EVQ}} + \delta \rho_F$.

Explicit form of $\gamma(b)$. The Fisher utility in the Euler equation $\alpha \rho + \beta(G\rho) - \lambda_F V_b + \nu = 0$ has $V_b(\phi) = \eta_F(\phi) b^{-2\phi}$, where $\eta_F(\phi)$ is the activation-conditioned Fisher coefficient with

$$\eta_F(\phi_k) = \frac{1}{2s_{\text{att}}^2} \mathbb{E}_{\ell, h, i, j} [w_{\ell hij} |\alpha_{\ell hij, k}|^2], \quad \alpha_{\ell hij, k} = q_{i, k}^{\mathbb{C}} \overline{k_{j, k}^{\mathbb{C}}}. \quad (44)$$

Using $(G\rho)'' = -\rho$, differentiating the Euler equation twice gives $\rho'' - \tau^2 \rho = (\lambda_F/\alpha)V_b''$. For slowly varying η_F compared with $e^{-s\phi}$,

$$\gamma(b) = \frac{\lambda_F}{\alpha} \bar{\eta}_F (2 \log b)^2 + O\left(\frac{|\eta_F'|}{s\bar{\eta}_F} + \frac{|\eta_F''|}{s^2 \bar{\eta}_F}\right). \quad (45)$$

The full $\eta_F(\phi)$ -local version is $\gamma_{\text{loc}}(\phi; b) = (\lambda_F/\alpha)[\eta_F''(\phi) - 2s\eta_F'(\phi) + s^2\eta_F(\phi)]$.

Suppression is L^1 /CDF, not pointwise. The particular solution $\delta \rho_F^{\text{part}} = \gamma(b) q_s/(s^2 - \tau^2)$ combined with (45) gives pointwise amplitude

$$\delta \rho_F^{\text{part}}(\phi) \approx \frac{\lambda_F \bar{\eta}_F}{\alpha} \frac{s^2}{s^2 - \tau^2} e^{-s\phi}, \quad (46)$$

so the $(s^2 - \tau^2)^{-1}$ factor is cancelled by the s^2 in $\gamma(b)$, and $\delta \rho_F^{\text{part}}$ is *not* pointwise small in s . The true suppression comes from the boundary-layer width of q_s : $\|q_s\|_1 = (1 - e^{-s})/s = O(1/\log b)$ and $\|q_s\|_2 = O(1/\sqrt{\log b})$. Hence

$$\|\rho_{\text{full}} - \rho_{\text{EVQ}}\|_1 \leq C_{\text{bc}}^{(1)}(\tau) \frac{\lambda_F \bar{\eta}_F}{\alpha} \frac{s^2}{s^2 - \tau^2} \frac{1 - e^{-s}}{s} = O\left(\frac{1}{\log b}\right), \quad (47)$$

$$\|\rho_{\text{full}} - \rho_{\text{EVQ}}\|_2 \leq C_{\text{bc}}^{(2)}(\tau) \frac{\lambda_F \bar{\eta}_F}{\alpha} \frac{s^2}{s^2 - \tau^2} \sqrt{\frac{1 - e^{-2s}}{2s}} = O\left(\frac{1}{\sqrt{\log b}}\right), \quad (48)$$

and the CDF-level error is $\|F_{\text{full}} - F_{\text{EVQ}}\|_\infty \leq \|\delta \rho_F\|_1 = O(1/\log b)$. Fisher forcing is therefore globally small in CDF / L^1 transport at large b , but not necessarily pointwise small near $\phi=0$.

Warp amplification at large τ . Via the inverse-CDF map,

$$\|\phi_{\text{full}} - \phi_{\text{EVQ}}\|_\infty \leq \frac{\|\delta \rho_F\|_1}{\inf_\phi \rho_{\text{full}}(\phi)}, \quad \inf_\phi \rho_{\text{EVQ}}(\phi) = \frac{\tau}{\sinh \tau}, \quad (49)$$

so CDF perturbations are amplified by $\sinh \tau/\tau$ when translated into warp errors; very short L or very large d_{head} should re-include the forcing branch.

881 **Dimensionless diagnostic.** Using $\|\rho_{\text{EVQ}} - 1\|_2 = \tau^2/\sqrt{45} + O(\tau^4)$ and $\Gamma_b :=$
 882 $(\lambda_F \bar{\eta}_F / \alpha) s^2 / ((s^2 - \tau^2)\tau^2)$, the L^2 -level ratio is

$$R_F := \frac{\|\rho_{\text{full}} - \rho_{\text{EVQ}}\|_2}{\|\rho_{\text{EVQ}} - 1\|_2} \leq C_{\text{bc}}^{(2)} \Gamma_b \sqrt{45} \sqrt{\frac{1-e^{-2s}}{2s}} = O\left(\frac{1}{\sqrt{\log b}}\right). \quad (50)$$

883 **Second-order utility deficit.** Writing the surrogate objective as a quadratic form with operator
 884 A_τ , the full branch improves the objective by

$$\Delta_F = \frac{1}{2} \gamma(b)^2 \langle q_s^\perp, A_\tau^{-1} q_s^\perp \rangle, \quad (51)$$

885 so dropping the forcing term is $O(\gamma(b)^2)$: if $R_F \leq r$, the missed correction is $O(r^2)$ relative to the
 886 tether deformation energy.

887 **Practical rule.** Retain the forcing branch whenever the measured R_F exceeds a tolerance r_{tol}
 888 (we use 0.05 for training, 0.01 for theory plots), or when $\tau/(2 \log b)$ is not small. The numerical
 889 threshold depends on the paper’s Fisher normalization $\lambda_F \bar{\eta}_F / \alpha$, which we treat as a measurable
 890 constant rather than a derived value.

891 A.17 Channel-load motivation for Pearson χ^2

892 **Normalization convention.** Throughout this paper, $S_{\chi^2}(\tau)$ denotes the d_{head} -normalized stiffness
 893 $S_{\chi^2}(\tau) := \frac{1}{d_{\text{head}}} \int_0^1 (1 - \rho_\tau)^2 / \rho_\tau d\phi$, with leading expansion $\tau^4 / (45 d_{\text{head}}) + O(\tau^6)$ consistent with
 894 Proposition 1. Identities below that produce a bare $\tau^4/45$ leading coefficient apply to $d_{\text{head}} S_{\chi^2}(\tau)$
 895 (the un-normalized integral); the d_{head} factor enters via the surrogate diagonal $\alpha = 1/d_{\text{rot}} (= 1/d_{\text{head}}$
 896 for MHA via §A.11; $= 1/d_{\text{rot}}$ for compressed-RoPE MLA via §C.1).

897 **Channel-load axioms.** Inverse-CDF quantization places channels with cell width $\ell(\phi) = 1/\rho(\phi)$
 898 and samples channels according to $dP_\rho = \rho d\phi$. Under the modeling axioms (i) channel-measure
 899 averaging, (ii) load ℓ as the primitive local quantity, (iii) second central moment, (iv) uniform density
 900 has zero stiffness, (v) local additivity, the canonical load-variance penalty is

$$S(\rho) = \text{Var}_{\phi \sim \rho d\phi} \left[\frac{1}{\rho(\phi)} \right] = \int_0^1 \frac{(1 - \rho(\phi))^2}{\rho(\phi)} d\phi = S_{\chi^2}(\rho). \quad (52)$$

901 We treat this as a canonical motivated choice under the stated axioms, not a uniqueness claim.

902 **Load-moment exponent.** Distinct from the weighted- L^2 family $S_p(\tau) = \frac{1}{d_{\text{head}}} \int (\rho_\tau - 1)^2 / \rho_\tau^p d\phi$
 903 used in Table 8 (where Pearson χ^2 is $p=1$), we also examine a load-moment family
 904 $S_p^{\text{mom}}(\rho) = \mathbb{E}_\rho[(1/\rho - 1)^p]$. Substituting the EVQ cosh density and minimizing $S_p^{\text{mom}} - \lambda U$ gives
 905 $\tau^{2(p-1)} \asymp d_{\text{head}}^2 / L_{\text{eff}}^J$, hence $\tau^* \propto (L_{\text{eff}}^J)^{-1/(2p-2)}$; only $p=2$ yields the observed $L^{-1/2}$ exponent
 906 within this family. The two families parametrize different f -divergence neighborhoods and need not
 907 match on the finite- τ grid.

908 **f -divergences at finite τ .** Smooth f -divergences (KL, Jeffreys, Rényi) are locally quadratic
 909 around $\rho=1$ and share the leading $-1/2$ exponent asymptotically. Their *finite- τ* best points over
 910 the $L \in [128, 4096]$ grid used in Table 8 fit exponents in $[0.58, 0.81]$, outside the $\pm 1\%$ PPL basin
 911 around $\gamma=0.500$ at the tested scales.

912 A.18 λ as a curvature ratio

913 **Invariant definition.** Fix the normalizations of U_{unit} and U_{exact} in the $\theta = \tau^2$ expansion. Define

$$\lambda_\infty := \frac{\partial_\theta U_{\text{exact}}(\rho_{\sqrt{\theta}}; L, b)|_{\theta=0}}{\partial_\theta U_{\text{unit}}(\rho_{\sqrt{\theta}}; L, b)|_{\theta=0}} = \frac{u_2^{\text{exact}}}{u_2^{\text{unit}}}. \quad (53)$$

914 Writing $U_{\text{exact}} = u_2^{\text{exact}}(L, b) (d_{\text{head}} / L_{\text{eff}}^J) \tau^2 + O(\tau^4)$ makes λ_∞ the utility-curvature
 915 ratio absorbed by the unit convention. The stiffness side remains $d_{\text{head}} S_{\chi^2}(\tau) =$
 916 $(\sinh \tau / \tau^2) \arctan(\sinh \tau) - 1 = \tau^4/45 + O(\tau^6)$ (cf. “Normalization convention” in §A.17).

A posteriori consistency check. Under the $\lambda=1$ unit convention, the surrogate prediction is $c_{\text{pred}}(L, b) = \sqrt{45 Q_1(L, b)}$. Direct numerical integration of $Q_1(L, b) = \int_0^1 a(\phi) q(Lb^{-\phi}) d\phi$ at $b=500\text{K}$, $L \in \{256, 512, 1024\}$ gives $Q_1 \in [0.0314, 0.0319]$ (near-constant on this grid), hence $c_{\text{pred}} \approx 1.19$. The exact-kernel collision minimizer $c_{\text{coll}}=1.171 \pm 0.001$ (9 configurations, Table 6) agrees with this to within 2% (mean $c_{\text{coll}}/c_{\text{pred}}=0.981 \pm 0.003$, CV 0.28%), so $(c_{\text{coll}}/c_{\text{pred}})^2 = 0.96 \pm 0.005$ is within $\sim 4\%$ of unity. This is an *a posteriori* consistency check for the $\lambda=1$ convention inside the flat empirical basin; we do not treat λ as an independently derived physical constant.

Validation protocol (further). A tighter check would compute $u_2^{\text{exact}} = \partial_{\tau^2} U_{\text{exact}}(\rho_{\tau}; L, b)|_{\tau=0}$ directly from the oscillatory RoPE kernel and compare against $u_2^{\text{unit}} = Q_1(L, b)$; the collision-score proxy above suggests the ratio is already within 4% of unity across our configurations.

A.19 Discrete-channel transport gap and the MLA channel-scarcity mechanism

Setup. Inverse-CDF quantization replaces the continuous density ρ by an atomic measure. Let $F(\phi) = \int_0^{\phi} \rho$, $Q(u) = F^{-1}(u)$, midpoint grid $u_k = (k - \frac{1}{2})/K$, and $\phi_k = Q(u_k)$. The empirical channel measure is $\mu_K = K^{-1} \sum_{k=1}^K \delta_{\phi_k}$. Direct $\|\rho_K - \rho\|$ is ill-posed: one cannot subtract an atomic measure from a density. We therefore work in Wasserstein distance for transport and in a quantile-cell histogram ρ_K for density.

Transport bounds. Assume $0 < m \leq \rho \leq M$ and $\|\rho'\|_{\infty} \leq B$; then Q is $(1/m)$ -Lipschitz. The midpoint quantization errors satisfy

$$W_{\infty}(\mu_K, \mu_{\rho}) \leq \frac{1}{2Km}, \quad W_1(\mu_K, \mu_{\rho}) \leq \frac{1}{4Km}. \quad (54)$$

Defining the quantile-cell histogram $\rho_K(\phi) = 1/(K|I_k|)$ on $I_k = [Q((k-1)/K), Q(k/K)]$, the histogram density satisfies

$$\|\rho_K - \rho\|_1 \leq \frac{B}{Km}, \quad \|\rho_K - \rho\|_{\infty} \leq \frac{B}{Km}. \quad (55)$$

Application to EVQ-Cosh. For $\rho_{\tau}(\phi) = \tau \cosh(\tau(1-\phi))/\sinh \tau$, $m_{\tau} = \tau/\sinh \tau$ and $B_{\tau} = \tau^2$. Substituting,

$$W_1(\mu_{K,\tau}, \mu_{\rho_{\tau}}) \leq \frac{\sinh \tau}{4K\tau}, \quad \|\rho_{K,\tau} - \rho_{\tau}\|_1 \leq \frac{\tau \sinh \tau}{K} = \frac{\tau^2}{K} + O\left(\frac{\tau^4}{K}\right). \quad (56)$$

Kernel-integrated errors. For a smooth two-variable kernel K_{sm} with Lipschitz constant L_K in each argument,

$$\left| \iint K_{\text{sm}} d\mu_K d\mu_K - \iint K_{\text{sm}} d\mu_{\rho} d\mu_{\rho} \right| \leq \frac{L_K}{2Km} + O(K^{-2}). \quad (57)$$

The singular δ -component of K_{app} must be interpreted as a cell average (not as a literal atom-kernel pairing), after which the bound applies to the smoothed kernel.

High-resolution distortion. For a smooth weighted distortion w ,

$$\mathcal{D}_K[\rho] \approx \frac{1}{12K^2} \int_0^1 \frac{w(\phi)}{\rho(\phi)^2} d\phi + O(K^{-3}). \quad (58)$$

This is the finite-channel mechanism: a shaped ρ reduces the weighted inverse-density load $\int w/\rho^2$; the absolute benefit scales as K^{-2} for smooth quantization error and as K^{-1} for transport/kernel terms.

Channel scarcity amplifies the density-shape correction. Going from standard MHA ($K=64$) to MLA ($K=16$, i.e. $d_{\text{rope}}=32$) scales the K^{-2} component of the distortion bound by a factor of 16 and the K^{-1} component by a factor of 4. The observed PPL-gain ratio between MLA (31.1%

at $K=16$; §4.4) and MHA (-13.3% at $K=64$, 3-seed 454M FineWeb-Edu row in Table 21; with 1-seed rows ranging -10.9% to -18.9%) is approximately $2\times$: this is directionally consistent with the predicted K^{-1}/K^{-2} amplification but is not a quantitative match, since the realized PPL gain is moderated by architecture, optimization dynamics, and loss nonlinearities that the surrogate does not model. The bounds refine the qualitative observation that channel scarcity amplifies shape corrections into explicit K^{-1}/K^{-2} scaling on the surrogate distortion.

Caveat. At large τ (short L or large d_{head}), $m_\tau = \tau / \sinh \tau$ becomes small and the $1/(Km_\tau)$ factor grows, so these bounds become vacuous in the very-large- τ regime where the pure-tether branch itself needs re-inclusion of the forcing term (§A.16).

B Experimental Details

B.1 Primary body experiments

The body centers three empirical anchors:

1. the EVQ $\times$ YaRN systems result on passkey-mix data,
2. DAPE-style PE-dominant extrapolation at 128 tokens,
3. the 432M MLA scarce-channel stress test with $d_{\text{rope}} = 32$.

These anchors were selected because they directly stress the substrate-range composition, PE-dominant allocation, and scarce-channel regimes emphasized in the main text. The $L = 256$ raw/YaRN curves are retained as supporting confirmation rather than a primary anchor.

B.2 Reproducibility snapshot

Table 10: Compact reproducibility summary for the primary experiments discussed in the main text.

Anchor	Train length	Scale	Seeds	Main datasets
EVQ $\times$ YaRN	2048	454M	3 + 3	FineWeb-Edu + 10% passkey mix
DAPE-style	128	125M	1–3	FineWeb-Edu
MLA scarce-channel	8192	432M	3	FineWeb-Edu / MLA setting

Table 11 lists the training hyperparameters shared across text experiments; per-tier deviations (learning rate, batch size) are noted.

Across all three anchors, the architectural intervention is deliberately minimal: model architecture, optimizer family, and training loop are kept fixed, and only the RoPE inverse-frequency initialization is changed. The principal base in the body is $b = 500\text{K}$, which is also the regime where the collision-block analysis predicts the largest headroom for non-geometric allocations.

B.3 Worked schedule and practitioner decision guide

For a standard MHA example with $d_{\text{head}}=64$, $K=32$, $L=2048$, and $b=500\text{K}$, the bare rule gives $\tau=1.414$. Midpoint quantization then gives the following selected entries (the full table is generated by `scripts/lib/rope/schedules.py`):

The current evidence supports a conditional decision guide rather than a universal recipe. For standard MHA in the tested $b \geq 500\text{K}$ regime, use the bare $d_{\text{head}}/\sqrt{L}$ rule as a basin selector. At smaller b and larger L , compute the explicit $c_{\text{pred}}(L, b) = \sqrt{45Q_1(L, b)}$ correction and treat it as analysis guidance until trained-text validation is available. For compressed-RoPE MLA, use $K=d_{\text{rot}}/2$ for quantization; the reported $d_{\text{eff}}=d_{\text{head}}$ choice is calibrated, and the $d_{\text{rope}}/\sqrt{L}$ ablation is still required. Finally, evaluate the Fisher-forcing diagnostic from Appendix A.16; if $R_F > 0.05$ or $\tau/(2 \log b)$ is not small, the pure-tether branch should not be treated as sufficient.

Table 11: Training hyperparameters for the primary text experiments. All tiers share the same optimizer and schedule; per-tier values are noted where they differ.

Parameter	Value
Optimizer	AdamW ($\beta_1=0.9$, $\beta_2=0.95$, $\epsilon=10^{-8}$)
Learning rate	6×10^{-4} (50M/125M), 2×10^{-4} (454M and 750M continue@4K)
LR schedule	Cosine decay to $0.1 \times$ peak LR
Warmup	2% of total steps (linear)
Batch size (seqs)	32 (50M), 16 (125M), 2–4 (454M/750M)
Weight decay	0.1
Gradient clipping	max-norm 1.0
Dropout	0.0 (no dropout)
Sequence packing	Contiguous, no cross-document masking
<i>Video DiT (129M–382M)</i>	
Optimizer	AdamW ($\beta_1=0.9$, $\beta_2=0.95$)
RoPE base	10,000 (temporal axis)
Activation	GELU (tanh approx.)
<i>MLA (432M)</i>	
LR / warmup	6×10^{-4} / 100 steps
Training tokens	500M

Table 12: Selected entries from the worked EVQ-Cosh schedule ($K=32$, $\tau=1.414$, $b=500K$).

Channel k	ϕ_k	$\omega_k = b^{-\phi_k}$
0	0.009877	0.8784348
7	0.162463	0.1186125
15	0.377917	0.0070188
23	0.650926	0.0001952
31	0.978625	0.0000026

987 B.4 Larger-scale supporting evidence

988 The 750M ‘continue@4K’ experiment is reported as supporting evidence only. The reason is not
989 that the result is weak; on the contrary, it shows 16K PPL improving from 45.1 to 24.4 and 8K AR
990 exact improving from 0% to 77.5%. The reason is evidence hygiene: the run is single-seed, and
991 LongBench is outside the controlled mechanism scope and is not used as a paper claim.

992 This distinction is important for interpretation. The larger-scale run is best viewed as evidence that
993 the same EVQ pattern persists after continued pretraining at a longer training length, not as the main
994 fairness anchor of the paper. The main fairness anchors remain the multi-seed 454M comparisons
995 used in the main text.

996 B.5 Progressive training dynamics

997 Table 14 and Figure 5 detail the supporting progressive training result referenced in the body. This is
998 a single-seed run separate from the primary EVQ×YaRN anchor above, with Stage 1 training length
999 $L=512$ before progressive extension. The key observation is the divergence pattern: EVQ+YaRN
1000 PPL remains flat across all three stages while Geo+YaRN degrades sharply at Stage 3, producing
1001 the widening relative advantage ($-34.6\% \rightarrow -52.0\% \rightarrow -81.2\%$). In this single-seed setup the gap
1002 is consistent with being driven by geometric RoPE’s progressive degradation rather than by further
1003 intrinsic EVQ improvement.

1004 B.6 Modality-dependent τ correction for video DiT

1005 The scaling law $\tau^* = d_{\text{head}}/\sqrt{L}$ matches text autoregressive models with causal attention. For
1006 bidirectional DiT, the best observed sweep point for τ is approximately $0.53 \times$ the text prediction

Table 13: Single-seed larger-scale consistency check: 750M continued pretraining from 2K to 4K. The same short-range-cost / long-range-gain pattern persists, and once retrieval saturates the sharper separation appears in AR exact.

Metric	Geo	EVQ
PPL@2K	25.9	26.2
PPL@4K	22.0	22.3
PPL@8K	23.4	19.6
PPL@16K	45.1	24.4
Passkey retrieval @8K	100%	100%
AR exact @8K	0%	77.5%
Global AR exact	66.67%	92.5%

Table 14: Progressive training widens EVQ’s advantage (454M, single seed, PPL@16K with YaRN).

Stage	L_{train}	Geo+YaRN	EVQ+YaRN	Δ
1	512	3.80	2.48	−34.6%
2	512→1024	4.60	2.21	−52.0%
3	512→1024→2048	13.17	2.48	−81.2%
Stage 3: PPL@48K		14.22	2.63	−82%

1007 ($\tau=1.5$ vs. predicted 2.83 for $K_t=16, T=32$). A schedule-aware decomposition writes the ratio as

$$\frac{\tau_{\text{video}}}{\tau_{\text{causal}}} \approx \frac{1}{\sqrt{2}} \sqrt{\mathbb{E}_t[\alpha_t^m]}, \quad (59)$$

1008 where $1/\sqrt{2}$ is the bidirectional softmax-Jacobian factor (a diffuse bidirectional window of size
1009 $2L+1$ halves the directional curvature relative to a causal window of size L ; Appendix A.15), and
1010 $\sqrt{\mathbb{E}_t[\alpha_t^m]}$ is a schedule-dependent signal-amplitude contraction with m determined by the utility
1011 definition. Our training run uses a Rectified-Flow (RF) schedule $\alpha_t=1-t$ with uniform $t \sim \mathcal{U}[0, 1]$
1012 and no loss weighting (not a VP SDE). Under RF, $\mathbb{E}_t[\alpha_t]=0.5$, $\mathbb{E}_t[\alpha_t^2]=1/3$, and $\mathbb{E}_t[\alpha_t^4]=0.2$, giving
1013 predicted ratios 0.500, 0.408, and 0.316 for $m \in \{1, 2, 4\}$ respectively. Only $m=1$ (linear signal-
1014 expectation utility) matches the empirical 0.53 within 6%; the covariance-linear $m=2$ is 23% below
1015 observation. We therefore report (59) as a directional decomposition and leave a principled utility-
1016 exponent derivation under RF (specifying why $m=1$ rather than $m=2$ is the correct contraction for
1017 the RoPE phase-coherence utility) as future work.

1018 B.7 Cross-modal evidence: video DiT with bidirectional attention

1019 To test whether EVQ transfers beyond autoregressive decoders, we evaluate on a 129.6M video diffu-
1020 sion transformer (DiT) with bidirectional attention and 3D RoPE. We compare EVQ-Cosh ($\tau=1.5$)
1021 against geometric allocation ($\tau=0$) on Oscillating Moving MNIST [Srivastava et al., 2015], training
1022 on 32 frames and extrapolating to 128 frames ($4\times$) with temporal YaRN. Crucially, both alloca-
1023 tions are trained *within the same run* (head-to-head), sharing identical weights, optimizer state, and
1024 CUDA non-determinism—controlling for shared optimizer state and CUDA nondeterminism and
1025 isolating the implemented frequency-allocation difference within this setup.

1026 Table 15 reports results across two independent seeds. EVQ wins every metric in both seeds. Aver-
1027 aging across seeds, Train MSE improves by -26% , all-frame extrapolation by -15% , and far-frame
1028 extrapolation by -32% . The consistency across both seeds—EVQ wins all three metrics in each—
1029 suggests that the advantage is not a stochastic artifact, though we note the sample size (two seeds)
1030 is too small for formal statistical testing.

1031 This supporting result probes EVQ beyond the autoregressive text setting along two axes simultane-
1032 ously: architecture (bidirectional attention, no causal mask) and modality (video diffusion). The
1033 shared-run head-to-head design controls major optimizer-state and CUDA nondeterminism con-
1034 founders within this setup, isolating the implemented frequency-allocation contrast more tightly
1035 than separate runs. A 382M DiT scale-up ($3\times$ larger, single seed) yields directionally consistent
1036 results (-35% far-frame MSE), and EVQ outperforms Geo under all three inference-time methods

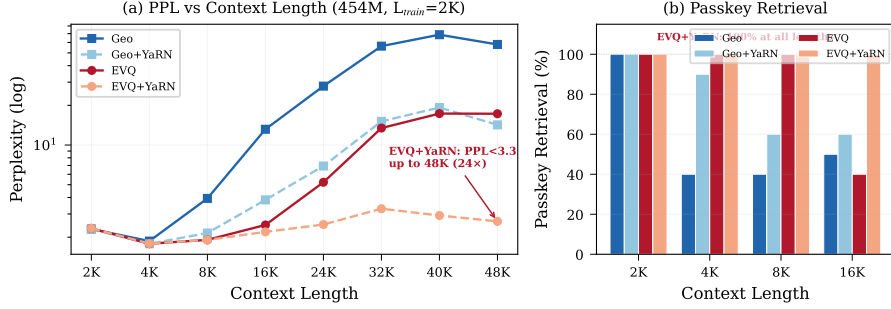


Figure 5: **Progressive training widens EVQ’s advantage.** Three-stage context extension (512 → 1024 → 2048, 454M, single seed). EVQ+YaRN remains stable across stages while Geo+YaRN degrades sharply, extending functional context to 48K (24× training length) at PPL 2.63.

Table 15: DiT head-to-head evaluation (129.6M, bidirectional attention, 3D RoPE). Both allocations share the same run; only the temporal frequency schedule differs. EVQ wins all metrics across both seeds.

Seed	Metric (YaRN)	Geo ($\tau=0$)	EVQ ($\tau=1.5$)	Δ
42	Train MSE	0.01350	0.01069	−21%
42	All extrap	0.00670	0.00573	−14%
42	Far extrap	0.00774	0.00502	−35%
137	Train MSE	0.01113	0.00761	−32%
137	All extrap	0.00713	0.00598	−16%
137	Far extrap	0.00856	0.00601	−30%
Mean Δ		−26% / −15% / −32%		

1037 tested (YaRN, RIFLEx, raw). This is reported as supporting evidence only; multi-seed replication is
 1038 left to future work at this scale.

1039 **Dead-channel mechanism: base reduction eliminates the phase transition.** At base = 10,000
 1040 with $K_t=16$ temporal frequencies and 32 training frames, the lowest-frequency channels accumu-
 1041 late less than 0.1 rad of phase rotation over the entire training sequence—they are effectively dead.
 1042 EVQ’s τ parameter activates these channels by redistributing frequency mass, but at base = 10,000
 1043 the activation is discontinuous: $\tau=1.2$ leaves 11/16 channels alive (effective rank 9), while $\tau=1.5$
 1044 activates 12/16 (rank 10), producing the sharp phase transition observed in Table 15.

1045 To test this mechanism, we repeat the head-to-head experiment with base = 1,000 (same architec-
 1046 ture, same run; 2 seeds). At this base, the geometric allocation already starts with 13/16 alive chan-
 1047 nels (rank 10), so the dead-channel bottleneck is resolved. Table 16 is consistent with the prediction:
 1048 (1) $\tau=1.2$ fully recovers, winning Geo by −48% far-frame MSE (vs. +162% at base = 10,000);
 1049 (2) $\tau=1.2$ and $\tau=1.5$ are statistically indistinguishable ($< 0.3\%$ difference on all metrics), support-
 1050 ing the interpretation that the phase transition was driven by the dead-channel threshold rather than
 1051 by an intrinsic property of $\tau=1.2$.

Table 16: Dead-channel mechanism check: base = 1,000 DiT head-to-head (129.6M, seed 42, YaRN). The sharp phase transition between $\tau=1.2$ and $\tau=1.5$ observed at base = 10,000 is no longer present in this seed; both EVQ configurations win Geo by $\approx 48\%$ far-frame MSE.

τ	Train MSE	All extrap	Far extrap	$\Delta_{\text{far vs Geo}}$
0.0 (Geo)	0.02037	0.01274	0.01646	—
1.2	0.00933	0.00820	0.00859	−48%
1.5	0.00930	0.00822	0.00858	−48%

Dead temporal channels across video DiT architectures. The dead-channel problem is not specific to our experimental setup. Table 17 audits five major video DiT architectures, all of which inherit $\text{base} = 10,000$ from text transformers. For each model, we compute the per-channel phase rotation $\theta_k \cdot T_{\text{train}}$ at the training temporal length, where $\theta_k = \text{base}^{-(k+0.5)/K_t}$ is the k -th temporal frequency. Channels with phase < 0.1 rad are counted as dead.

Every video model wastes 32–50% of its temporal frequency channels, because video temporal sequences (13–51 latent tokens after VAE compression) are 80–315 $\times$ shorter than text sequences (≥ 4096 tokens) for which $\text{base} = 10,000$ was designed. The text reference (LLaMA-2) has zero dead channels, consistent with the dead-channel problem being specific to the sequence-length mismatch in the video temporal axis. EVQ-Cosh addresses this by redistributing frequency mass into the alive zone, and Table 16 shows that the effect persists in the tested $\text{base} = 1,000$ dead-channel regime; we do not claim base-independent dominance.

Table 17: Dead temporal channels in major video DiT architectures. All models inherit $\text{base} = 10,000$ from text transformers. Channels with phase rotation < 0.1 rad at the training temporal length are effectively dead. Video models waste 32–50% of temporal channels; the text reference (LLaMA-2) has zero dead channels.

Model	K_t	T_{train}	Dead / K_t	Dead%	Base oversize
CogVideoX-5B [Yang et al., 2024b]	8	13	4/8	50%	56 $\times$
Wan-2.1 1.3B [Wan Team, 2025]	22	21	9/22	41%	42 $\times$
Latte-1 [Ma et al., 2024]	12	16	5/12	42%	50 $\times$
Open-Sora 1.2 [Zheng et al., 2024b]	22	51	7/22	32%	17 $\times$
HunyuanVideo [Kong et al., 2024]	8	33	3/8	38%	21 $\times$
Our DiT	16	32	6/16	38%	26 $\times$
LLaMA-2 7B [Touvron et al., 2023] (text)	64	4096	0/64	0%	—

Base sweep: EVQ robustness across the base spectrum. We sweep the temporal RoPE base across four orders of magnitude (100–10,000) to characterize how EVQ interacts with the dead-channel severity. Both GEO ($\tau=0$) and EVQ ($\tau=1.5$) are evaluated head-to-head within the same run at each base, under both YaRN-scaled and raw (no scaling) 4 $\times$ temporal extrapolation.

Table 18: Base sweep: far-frame denoising MSE ($t=0.5$, 4 $\times$ extrapolation) as a function of temporal RoPE base. EVQ is remarkably stable across $\text{base} \geq 500$; GEO+YaRN exhibits non-monotonic behavior with a catastrophic spike at $\text{base} = 5,000$.

Base	With YaRN			Without YaRN			Dead ch.
	GEO	EVQ	Δ	GEO	EVQ	Δ	
100	0.0161	0.0189	+18%	0.296	0.356	+20%	0/16
500	0.0210	0.0207	−1%	0.224	0.146	−35%	~3/16
1000	0.0517	0.0178	−66%	0.286	0.132	−54%	~5/16
5000	0.0787	0.0210	−73%	0.172	0.111	−36%	~5/16
10000	0.0401	0.0137	−66%	0.107	0.048	−55%	6/16
50000	0.1895	0.0262	−86%	0.247	0.108	−56%	~9/16

Two key findings emerge in this 1–2 seed video-DiT setting. First, when all channels are alive ($\text{base} = 100$, 0/16 dead), GEO wins by $\approx 20\%$, consistent with EVQ’s advantage being specific to the dead-channel regime. Second, EVQ+YaRN is stable across $\text{base} \geq 500$ (1.9 $\times$ MSE range vs. GEO’s 12 $\times$), indicating robustness to the base hyperparameter within the tested grid. The best observed grid point is EVQ at $\text{base} = 10,000$ with YaRN (0.0137), better than GEO at any tested base. A pure training-length evaluation (32f, no extrapolation) shows all 12 configurations produce near-identical MSE ($< 6\%$ variation), supporting the interpretation that the advantage is an extrapolation phenomenon rather than a training-distribution effect.

C Supporting Results

C.1 Multi-head Latent Attention

To test whether EVQ’s benefit extends to Multi-head Latent Attention (MLA), the compressed-RoPE attention variant introduced in DeepSeek-V2 [DeepSeek-AI, 2024a], we train a 432M-parameter MLA Transformer from scratch. Our configuration uses $d_{\text{rope}}=32$ and base = 500K; production DeepSeek-V2/V3 [DeepSeek-AI, 2024b] uses $d_{\text{rope}}=64$ and base = 10K. The shared architectural property—a compressed RoPE subspace with far fewer frequency channels than standard MHA—is what makes allocation quality decisive. Only 16 frequency channels carry positional information, while the remaining kv_rank=256 dimensions carry content. (d_{rope} in MLA notation is the same quantity as d_{rot} in §3; we use the DeepSeek-style d_{rope} throughout this subsection.)

The model trains on FineWeb-Edu (500M tokens) at $L_{\text{train}}=8192$ with $b=500\text{K}$ across three seeds (42, 43, 88). EVQ uses $\tau=1.414$ (i.e., $d_{\text{head}}/\sqrt{L} = 128/\sqrt{8192}$).

Two distinct dimensions. It is important to distinguish two roles played by the head dimension. (i) The *kernel normalization* $\alpha=1/(2K)=1/d_{\text{rope}}$ controls the diagonal coefficient of K_{app} and depends only on the number of RoPE frequency pairs ($K=d_{\text{rope}}/2$); for MLA this gives $\alpha=1/32$. (ii) The *operating-point dimension* d_{eff} in $\tau^*=d_{\text{eff}}/\sqrt{L}$ enters through the post-softmax transport utility U , which weights all attention-logit components carried by the head—including the non-RoPE content channels mixed in via the latent projection. The variational $\alpha=1/d_{\text{rope}}$ derivation alone would give a much smaller deployment τ ; we deploy $d_{\text{eff}}=d_{\text{head}}=128$ as a calibrated convention reflecting the latent-projection coupling, not as a consequence of the broadband surrogate. Table 19 evaluates allocation sensitivity at this fixed convention but cannot identify the optimal convention. A direct $\tau=d_{\text{rope}}/\sqrt{L}$ ablation is therefore required to test it.

Table 19: MLA validation (432M, $d_{\text{rope}}=32$, 3-seed mean $\pm$ std). EVQ alone outperforms GEO+YaRN(s=4) at 16K and ties within printed precision at 24K (both 204.9). EVQ+YaRN has the lowest mean PPL in this tested grid.

Method	PPL@8K	PPL@16K	PPL@24K	PPL@32K
GEO	35.4 $\pm$ 0.9	138.8 $\pm$ 5.5	241.5 $\pm$ 2.6	323.7 $\pm$ 5.2
GEO+YaRN(s=4)	35.5 $\pm$ 0.9	117.9 $\pm$ 6.5	204.9 $\pm$ 6.9	278.5 $\pm$ 6.3
EVQ	35.8 $\pm$ 0.8	95.6$\pm$4.1	204.9 $\pm$ 14.7	291.6 $\pm$ 20.5
EVQ+YaRN(s=4)	35.8 $\pm$ 0.8	71.1$\pm$4.1	153.2$\pm$12.5	236.6$\pm$15.9

Table 19 shows the results. Three findings stand out. First, EVQ alone reduces PPL at 16K by 31.1% (138.8 $\rightarrow$ 95.6) with only +1.1% in-distribution cost—a larger relative gain than observed with standard MHA at comparable scales, consistent with the “fewer channels, each more precious” hypothesis. Second, EVQ alone outperforms GEO+YaRN(s=4) at 16K (−31.1% vs −15.1%); at 24K the two are tied within printed precision (204.9 each). Under matched YaRN scale in this setup, a training-time allocation change is thus comparable to or better than the inference-time rescaling baseline we tested, but we do not claim superiority over every possible inference-time method. Third, EVQ+YaRN has the lowest mean PPL at each length in this tested grid; the ranking EVQ+YaRN > EVQ > GEO+YaRN > GEO holds strictly at 16K, with YaRN providing larger marginal benefit on EVQ (−25.6%) than on GEO (−15.1%) at 16K, consistent with complementary mechanisms.

Training-progression analysis within this 500M-token run shows EVQ’s advantage is visible from early training and grows monotonically to −31.1% at completion, while the in-distribution cost simultaneously decreases from +1.4% to +1.1%. This rules out a transient early-checkpoint explanation within the reported primary protocol, but does not establish durability at larger token budgets; the separate single-seed 1B schedule-sensitivity row is scoped accordingly.

Table 20: Matched-scale YaRN leverage at $L = 256$; NTK-aware scaling is not complementary at large τ .

Method	YaRN gain @4K	YaRN gain @8K
Geo	$-4.5\% \pm 1.2$	$-3.1\% \pm 1.6$
EVQ $\tau = 2.0$	$-27.0\% \pm 0.1$	$-28.9\% \pm 1.3$
EVQ $\tau = 4.0$	$-32.5\% \pm 2.0$	$-40.7\% \pm 1.6$
<i>NTK-aware at 32$\times$: Geo 198.1, EVQ2 143.3, EVQ4 331.4</i>		

Table 21: Multi-scale raw length generalization. Positive in-range deltas remain small, while long-range gains are consistent across scales. The 750M ‘continue@4K’ row is reported as a single-seed consistency check.

Scale	Dataset	Seeds	$\Delta\text{PPL}@2\text{K}$	$\Delta\text{PPL}@16\text{K}$
50M	TinyStories	1	-0.3%	-10.9%
125M	TinyStories	1	-1.7%	-18.9%
454M	TinyStories (Hybrid)	1	-0.4%	-13.7%
454M	FineWeb-Edu	3	+0.4%	-13.3%
750M continue@4K	Mix setting	1	+1.2%	-45.9%

1113 C.2 Matched-scale YaRN leverage (supporting table)

1114 C.3 Multi-scale raw length generalization

1115 C.4 Attention distance redistribution

1116 Figure 6 illustrates the waterbed crossover in learned attention patterns of the 750M continue@4K
 1117 checkpoint, complementing the surrogate-level analysis in §A.8.

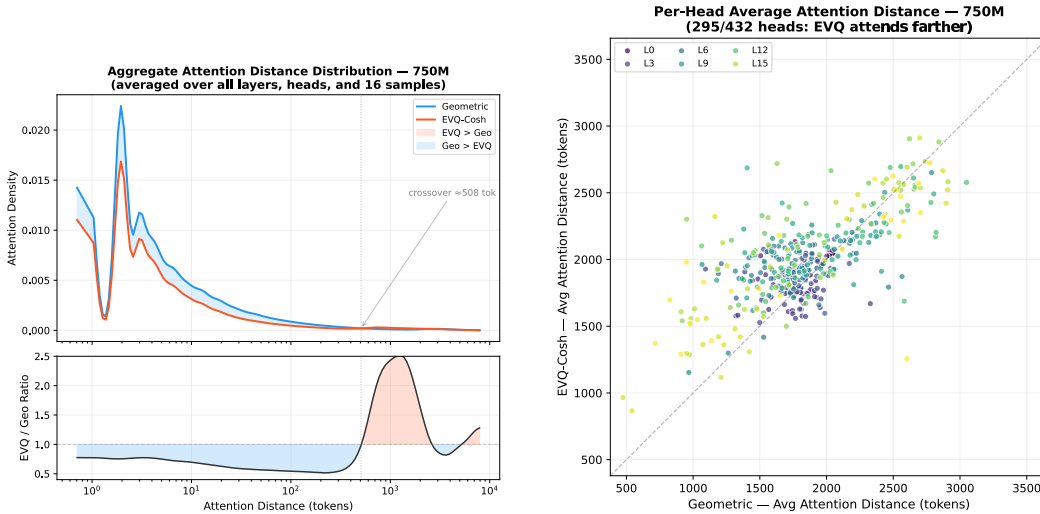


Figure 6: **Waterbed illustration in learned attention patterns (750M, single seed).** Left: aggregate attention density vs. token distance. A crossover at ≈ 508 tokens separates a short-range regime (Geometric dominant) from a long-range regime (EVQ dominant, up to $2.5\times$ higher density). Right: per-head average attention distance. 295/432 heads (68%) attend farther under EVQ, with the strongest shifts in deeper layers (L12–L17).

1118 C.5 Downstream QA evaluation: QuALITY Gold NLL

1119 We evaluate QuALITY [Pang et al., 2022] on 454M checkpoints initialized from the 2K pretraining
 1120 run, continued at 4K, and then finetuned on QuALITY at 4K. Thus 4K is in-distribution for this
 1121 downstream protocol, although the initial pretraining checkpoint used $L=2048$. Table 22 reports

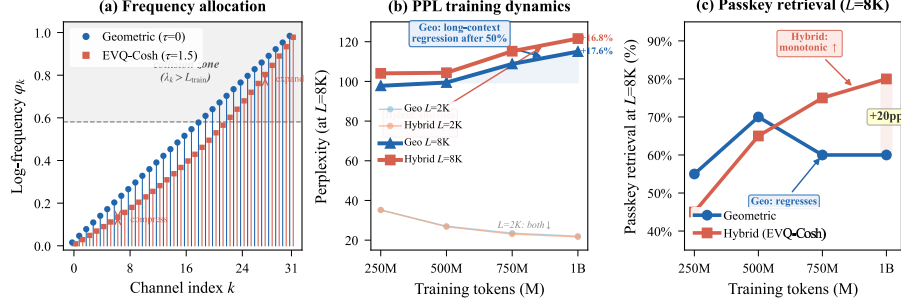


Figure 7: **750M frequency dynamics and retrieval divergence** (single seed). EVQ compresses the ultra-low-frequency/dead tail and reallocates usable resolution into the active spectral band, producing qualitatively different retrieval dynamics at long range.

1122 the clean gold-answer-NLL evaluation on all $n=2086$ validation examples with distractor padding
 1123 and random article position 20%–80%. At this model scale, 4-option accuracy hovers near the 25%
 1124 random baseline for all configurations, consistent with 454M capacity—not positional encoding—
 1125 being the binding constraint on downstream accuracy. However, gold-answer NLL separates the two
 1126 RoPE variants: EVQ’s NLL advantage grows from -1.7% at 4K (in-distribution) to -30.1% at 8K
 1127 ($2\times$ extrapolation) and -21.4% at 16K ($4\times$). YaRN at 8K recovers Geo’s accuracy to 4K level but
 1128 does not close the NLL gap (-8.1%). A separate earlier 200-sample accuracy-only diagnostic at
 1129 4K–32K is not part of Table 22 or Figure 8.

Table 22: QuALITY QA downstream evaluation (454M, $n=2086$, single seed). Accuracy is near random for all configs; gold-answer NLL reveals EVQ’s advantage in probability space.

Length	Mode	Accuracy (%)		Gold NLL ↓		NLL Δ
		Geo	EVQ	Geo	EVQ	
4K	raw	26.1	26.8	2.220	2.182	-1.7%
8K	raw	24.6	26.8	3.202	2.239	-30.1%
8K	YaRN	26.5	26.6	2.389	2.195	-8.1%
16K	raw	24.1	23.7	7.915	6.220	-21.4%

1130 Figure 8 visualizes the NLL divergence across context lengths, showing that EVQ’s advantage grows
 1131 with extrapolation ratio.

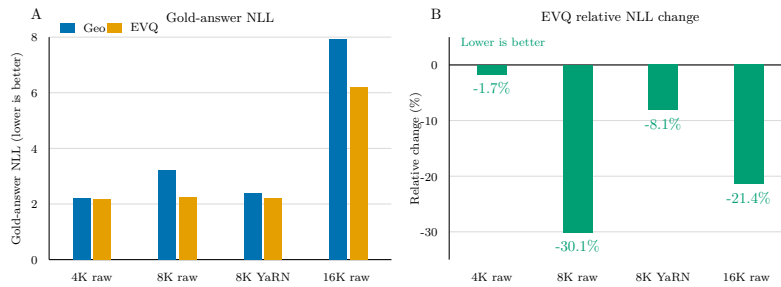


Figure 8: **Downstream QA: Gold-answer NLL across context lengths** (454M, QuALITY, $n=2086$). EVQ’s NLL advantage grows from -1.7% at in-distribution (4K) to -30.1% at $2\times$ extrapolation (8K).

1132 The key takeaway is that accuracy at the capacity floor is not a useful discriminator, but NLL on
 1133 the correct answer preserves the signal. This is consistent with the PPL and passkey evidence in
 1134 the body: EVQ improves the model’s probabilistic calibration at extrapolated lengths even when
 1135 accuracy is saturated by capacity constraints.

1136 C.6 Multi-scale waterbed verification

1137 Figure 9 visualizes the waterbed effect across model scales: EVQ trades a small in-range cost
 1138 ($\pm 1.7\%$) for long-range gains (10–46%) across the tested grid.

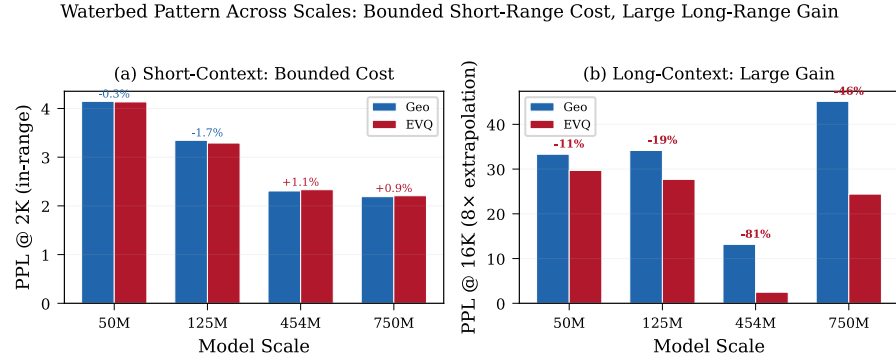


Figure 9: **Waterbed illustration across model scales.** EVQ pays a bounded in-range cost (left of dashed line) while achieving substantial long-range gains. The pattern is consistent from 50M to 750M within the tested grid, consistent with the theoretical prediction.

1139 D Supporting Experiments (from Main Text)

1140 The following results extend EVQ beyond the primary MHA setting. Each is scoped to a specific
 1141 question and labeled with its seed count.

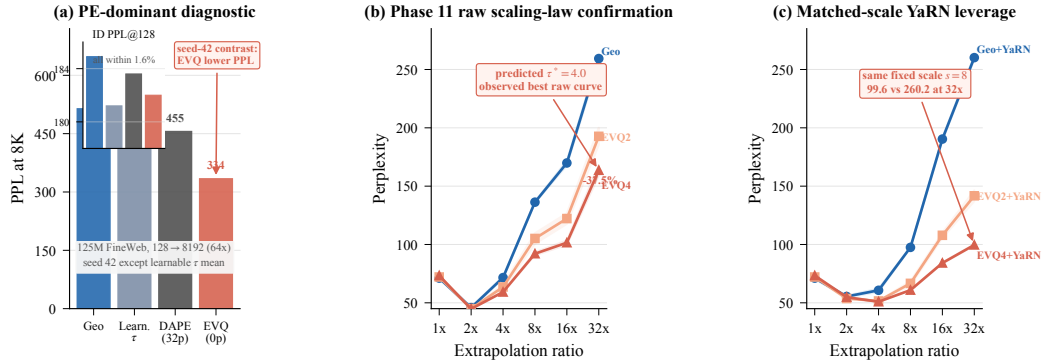


Figure 10: PE-dominant evidence (visualization of Table 4). Left: in the seed-42 128→8K contrast, EVQ’s closed-form allocation (zero learned parameters) attains lower extrapolation PPL than the learned positional-operator baseline. Right: at $L=256$, 3-seed mean curves show that the predicted $\tau^*=4.0$ is favored among tested settings, and EVQ4+YaRN outperforms Geo+YaRN across extrapolation ratios.

Table 23: $L=256$ YaRN composition study (125M, 3-seed mean PPL). The predicted $\tau^*=4.0$ is the best of the tested settings, and EVQ4+YaRN outperforms Geo+YaRN at every extrapolation ratio. This supplements Table 4 in the main text.

Method ($L=256$)	4×	8×	16×	32×
Geo+YaRN	60.7	97.5	190.4	260.2
EVQ2+YaRN	51.6	66.7	107.9	141.7
EVQ4+YaRN	50.9	60.9	84.1	99.6

MLA: allocation matters more when channels are scarce. EVQ extends to MLA [DeepSeek-AI, 2024a,b], where only 16 frequency channels carry positional information. On a 432M model (3-seed, 500M tokens, $L_{\text{train}}=8192$), EVQ alone reduces $2\times$ extrapolation PPL by 31.1% and outperforms Geo+YaRN at 16K. Supporting single-seed evidence under thorough training (1B tokens) suggests that EVQ’s raw advantage reverses (+11.1%) while EVQ+YaRN after fine-tuning still wins (−2.5%)—a 13.6pp swing consistent with the possibility that the frequency substrate remains a constraint even when raw extrapolation saturates. This row is not used as a primary claim. Full 3-seed standalone results are in Appendix C.1.

Post-training injection via LoRA (8B scale). As supporting post-hoc evidence, we inject EVQ-Cosh frequencies ($\tau=1.414$, the τ^* predicted for $d_{\text{head}}=128$, $L=8192$) into LLaMA-3-8B-Instruct [Grattafiori et al., 2024] and fine-tune with LoRA ($r=64=K$, $\alpha=128$) on LongAlign-10k [Bai et al., 2024a] for 300 steps.

Table 24: EVQ-Cosh LoRA on LLaMA-3-8B-Instruct ($r=64$, 300 steps, single seed). EVQ trades a modest in-distribution cost for substantial extrapolation gains.

Context	Base PPL	EVQ-LoRA PPL	Δ
8K (in-dist.)	7.42	9.63	+30%
16K ($2\times$)	176.3	21.5	−88%
32K ($4\times$)	1942.5	104.3	−95%

Table 24 shows that EVQ-LoRA reduces extrapolation PPL by $8\times$ at 16K and $19\times$ at 32K with a modest in-distribution cost (+30% PPL at 8K), serving as supporting post-hoc evidence at 8B scale. The rank is set at $r=d_{\text{head}}/2=64$. The phenomenological variational model in §A.13 predicts a rank-dependent viability threshold near $r \approx d_{\text{head}}/2$ under this single-model calibration; it is a testable hypothesis, not general rank-selection guidance. Below the fitted threshold, coupling stiffness appears to prevent frequency redistribution; above it, extrapolation recovers. RULER scores do not improve, consistent with LoRA capacity sufficing for positional adaptation but not instruction-following at extrapolated lengths; a matched Geo+LoRA+LongAlign control on identical training setup is required to isolate the EVQ effect from LoRA capacity.

Progressive training (single seed, preliminary). Three-stage progressive training (454M, 512→1024→2048) is reported as supporting evidence. The EVQ+YaRN PPL gap at 16K widens from −34.6% to −81.2%, driven by Geo+YaRN degradation while EVQ+YaRN remains stable. At Stage 3, functional context reaches 48K ($24\times$ training length) at PPL 2.63. After Stage 2, EVQ *raw* (PPL 11.2 at 16K) surpasses EVQ+YaRN at Stage 1 (16.8), suggesting that continued training can partially internalize the allocation benefit. This single-seed row is not used as a primary claim.

Cross-modal mechanism check: video DiT. In 129.6M DiT head-to-head experiments (2 seeds, shared training run), EVQ wins every metric in both seeds, eliminating optimizer and CUDA confounds. Once the dead-channel bottleneck is removed (base-1000), both $\tau=1.2$ and $\tau=1.5$ beat Geo by $\approx 48\%$ on far-frame MSE. This is a supporting mechanism check rather than a standalone video contribution: the collision bottleneck identified by the theory appears across modalities. Full results are in §B.6.

Signal gradient across abstraction layers. Across the diagnostics reported here, EVQ’s effect generally attenuates through successive abstraction layers: raw PPL −46% (Table 21) → Gold-Answer NLL −30% (Table 22) → passkey +59pp vs. Geo baseline (EVQ+YaRN 100% vs. Geo 41%, 3 seeds). QuALITY 4-option accuracy is near its capacity floor and has no stable directional gain across the four table rows (+0.7, +2.2, +0.1, and −0.4 percentage points), so we do not use it as positive evidence. This pattern is consistent with the intervention being largest at the PE layer and attenuated through higher abstraction where model capacity becomes the binding constraint.

Capability preservation. Table 25 shows that EVQ is non-destructive when retrieval is absent from training and improves extrapolated retrieval by +10–13pp when retrieval is learned (454M, 3-seed).

Table 25: Capability preservation and retrieval robustness (454M, 3-seed, *per-document* PPL scoring; cf. Table 3 for full-sequence scoring on the same model). EVQ is non-destructive when retrieval is absent, improves extrapolated retrieval, and is more robust to denser passkey supervision. The absolute PPL@16K differs from Table 3 (253.2 vs. 262.0) because of the scoring protocol only; the relative Geo→EVQ ranking is unchanged.

Setting	Geo	EVQ	Interpretation
Pure FineWeb-Edu passkey	55.7%	56.7%	non-destructive
10% mix retrieval @2K	100%	100%	both learn task
10% mix retrieval @4K	58.7%	68.7%	+10.0pp
10% mix retrieval @8K	40.7%	53.3%	+12.7pp
10% mix global retrieval	66.7%	74.0%	+7.3pp
10% mix PPL@2K	67.2	67.9	+1.0%
10% mix PPL@8K	161.9	150.3	-7.2%
10% mix PPL@16K	262.0	237.2	-9.5%
5% → 10% delta @8K	-13.3pp	-3.3pp	EVQ more robust

1185 **Table 26:** Method-landscape comparison supporting the Related Work positioning. EVQ-Cosh is the row targeting closed-form variational
1186 training-time frequency allocation for standard RoPE.

Method	Train alloc.	Infer scale	Post-hoc/ search	Learned	Variational allocation	Main axis
Geometric RoPE [Su et al., 2024]	fixed	—	—	—	—	default operator
NTK / PI [Rozière et al., 2023, Chen et al., 2023]	—	✓	post-hoc	—	—	global range
YaRN [Peng et al., 2024]	—	✓	post-hoc	—	—	piecewise range
LongRoPE [Ding et al., 2024]	—	✓	search+adapt	—	—	progressive range
LongRoPE2 [Shang et al., 2025]	—	✓	search+adapt	—	—	per-channel range
DAPE [Zheng et al., 2024a]	—	—	training	✓	—	adaptive operator
FIRE [Li et al., 2024]	—	—	training	✓	—	learned function
VideoRoPE [Wei et al., 2025]	heuristic	✓	design	—	—	modality-specific
Resonance RoPE [Wang et al., 2024]	—	✓	design	—	—	critical frequencies
MHRoPE/HoPE [Huang et al., 2026, Li et al., 2025]	heuristic	—	design	—	—	VLM axis-split
CARoPE [Veisi et al., 2025]	—	—	training	✓	—	learned dynamic frequencies
CoPE [Li et al., 2026]	—	✓	post-hoc	—	—	low-frequency clipping
EVQ-Cosh (ours)	✓	—	—	—	✓	train-time allocation

1188 NeurIPS Paper Checklist

1189 1. Claims

- 1190 • **Question:** Do the main claims made in the abstract and introduction accurately reflect
1191 the paper’s contributions and scope?
- 1192 • **Answer:** [Yes]
- 1193 • **Justification:** The abstract and introduction are restricted to three contributions: (1)
1194 identifying training-time frequency allocation as a PE design axis with a closed-form
1195 variational family, (2) demonstrating that a semi-analytic rule—with derived d_{head}
1196 and $L^{-1/2}$ structure and an empirically calibrated $\mathcal{O}(1)$ scale—falls within the best
1197 observed sweep basin across the tested grid of 27 configurations (multi-seed anchors
1198 support the systems and MLA claims; the PE-dominant DAPE-style diagnostic is ex-
1199 plicitly seed-42 for Geo/DAPE/EVQ), and (3) the matched-scale systems result that
1200 EVQ increases YaRN leverage under a fixed YaRN scale (we do not claim dominance
1201 over every tuned YaRN baseline). The τ^* rule’s epistemic status—derived d_{head} and
1202 $L^{-1/2}$ structure under the stated small- τ scalings, channel-load-motivated χ^2 stiffness,
1203 and unit-convention λ —is stated explicitly in §3.7.

1204 2. Limitations

- 1205 • **Question:** Does the paper discuss the limitations of the work performed by the au-
1206 thors?
- 1207 • **Answer:** [Yes]
- 1208 • **Justification:** The body explicitly marks the broadband surrogate as the main ap-
1209 proximation, presents τ^* as a semi-analytic operating-point rule rather than a tuned
1210 hyperparameter, and demotes single-seed and video evidence to supporting status.

1211 3. Theory assumptions and proofs

- 1212 • **Question:** For each theoretical result, does the paper provide the full set of assump-
1213 tions and a complete (and correct) proof?
- 1214 • **Answer:** [Yes]
- 1215 • **Justification:** Theorems 1–2 and Proposition 1 are exact conditional on the broadband
1216 surrogate; the surrogate itself is supported by a functional check (24–92% collision-
1217 score reduction across 12 configurations) in Appendix A.6. The τ^* scaling law (Propo-
1218 sition 1) is semi-analytic: d_{head} dependence and the $L^{-1/2}$ exponent are derived, but
1219 one transport constant λ is calibrated empirically. This status is stated explicitly in
1220 §3.7 and the limitations section.

1221 4. Experimental result reproducibility

- 1222 • **Question:** Does the paper fully disclose all the information needed to reproduce the
1223 main experimental results of the paper to the extent that it affects the main claims
1224 and/or conclusions of the paper?
- 1225 • **Answer:** [Yes]
- 1226 • **Justification:** The body states the principal settings and the appendix records datasets,
1227 seeds, evaluation grouping, and the status of larger supporting runs.

1228 5. Open access to data and code

- 1229 • **Question:** Does the paper provide open access to the data and code?
- 1230 • **Answer:** [Yes]
- 1231 • **Justification:** We include an anonymous supplementary code archive containing the
1232 EVQ-Cosh inverse-frequency initializer, the collision-surrogate and τ^* -rule analyses,
1233 evaluation scripts, and configuration files for the primary experiments (EVQ×YaRN
1234 454M, PE-dominant 125M/454M, MLA 432M). Upstream datasets (FineWeb-Edu,
1235 TinyStories, QuALITY, RULER passkey/NIAH) are available from their respective
1236 public releases; our passkey-mix composition and data-prep scripts are in the archive.
1237 Model checkpoints and weights are not included to preserve double-blind review and
1238 to keep the archive within size limits. A reproducibility README documents com-
1239 mands, provenance status, and any local-result requirements for regenerating or audit-
1240 ing the primary-tier tables and figures.

1241	6. Experimental setting/details
1242	• Question: Does the paper specify all the training and test details?
1243	• Answer: [Yes]
1244	• Justification: Main settings are given in the body and additional details are placed in
1245	the appendix.
1246	7. Experiment statistical significance
1247	• Question: Does the paper report error bars or equivalent multi-run evidence?
1248	• Answer: [Yes]
1249	• Justification: The matched-scale EVQ×YaRN and MLA claims are anchored in 3-
1250	seed evidence. The PE-dominant DAPE-style result is intentionally reported as a
1251	seed-42 mechanism diagnostic for Geo/DAPE/EVQ, while single- and two-seed scope
1252	extensions are labeled as supporting or exploratory.
1253	8. Experiments compute resources
1254	• Question: For each experiment, does the paper provide sufficient information on the
1255	computer resources used?
1256	• Answer: [Yes]
1257	• Justification: The appendix records model scales, training token counts, context win-
1258	dows, batch sizes, and seed counts for every primary run. Primary experiments were
1259	conducted on an anonymous internal A100 / H100 accelerator cluster; total compute
1260	for the primary tier (454M EVQ×YaRN + 432M MLA + 125M/454M PE-dominant)
1261	is on the order of a few hundred accelerator-hours, with the 454M/432M anchors
1262	dominating. Supporting LoRA-on-8B and video-DiT rows each use at most tens of
1263	accelerator-hours. Exact wall-clock varies with cluster scheduling and is not used as
1264	evidence for the scientific comparisons.
1265	9. Code of ethics
1266	• Question: Does the research conducted in the paper conform to the NeurIPS Code of
1267	Ethics?
1268	• Answer: [Yes]
1269	• Justification: The work studies frequency allocation for rotary position embedding
1270	and does not involve human-subject experimentation or a deployed user-facing sys-
1271	tem.
1272	10. Broader impacts
1273	• Question: Does the paper discuss both potential positive societal impacts and negative
1274	societal impacts of the work performed?
1275	• Answer: [Yes]
1276	• Justification: The limitations discussion and checklist justifications state both the
1277	potential systems benefits of better long-context positional design and the risk of over-
1278	generalizing beyond the present evidence.
1279	11. Safeguards
1280	• Question: Does the paper describe safeguards that have been put in place for respon-
1281	sible release?
1282	• Answer: [N/A]
1283	• Justification: The submission does not release a deployed system or a safety-sensitive
1284	product.
1285	12. Licenses
1286	• Question: Are the creators or original owners of assets used in the paper properly
1287	credited?
1288	• Answer: [Yes]
1289	• Justification: External methods, benchmarks, and model families are cited in the
1290	references.
1291	13. New assets
1292	• Question: Are new assets introduced in the paper well documented?

1293 • **Answer:** [N/A]
1294 • **Justification:** The submission does not introduce a released dataset or benchmark
1295 asset.

1296 14. **Crowdsourcing and human subjects**

1297 • **Question:** For crowdsourcing experiments and research with human subjects, does
1298 the paper include the full text of instructions given to participants?

1299 • **Answer:** [N/A]
1300 • **Justification:** No crowdsourcing or human-subject research is involved.

1301 15. **IRB approvals**

1302 • **Question:** Does the paper describe potential risks incurred by study participants?
1303 • **Answer:** [N/A]
1304 • **Justification:** No human-subject study is involved.

1305 16. **Declaration of LLM usage**

1306 • **Question:** Does the paper describe the usage of LLMs if it is an important, original,
1307 or non-standard component of the core methods in this research?
1308 • **Answer:** [N/A]
1309 • **Justification:** LLMs and agents were used only for author-side writing, editing, code
1310 assistance, and submission-hygiene checks. They are not an important, original, or
1311 non-standard component of the EVQ-Cosh method, theoretical derivation, experi-
1312 ments, or evaluation protocol.